

Premier League Match Dataset

Exploratory Data Analysis Report

Total Matches:	3,800
Total Features:	55
Numerical Features:	50
Categorical Features:	4
Total Missing Values:	0
Home Wins (H):	1,691 (44.5%)
Draws (D):	886 (23.3%)
Away Wins (A):	1,223 (32.2%)

Numerical Feature Summary Statistics

	missing	missing_%	count	mean	std	min	25%	median	75%	max
HPPG	0.0000	0.0000	3800.0000	1.3384	0.6466	0.0000	0.9375	1.2941	1.7500	3.0000
HPPGH	0.0000	0.0000	3800.0000	1.4460	0.7606	0.0000	1.0000	1.4375	2.0000	3.0000
HPPG_FORM	0.0000	0.0000	3800.0000	1.3261	0.7679	0.0000	0.8000	1.4000	1.8000	3.0000
HPPGH_FORM	0.0000	0.0000	3800.0000	1.4690	0.8349	0.0000	0.8000	1.4000	2.0000	3.0000
APPG	0.0000	0.0000	3800.0000	1.3494	0.6437	0.0000	0.9630	1.3125	1.7500	3.0000
APPGA	0.0000	0.0000	3800.0000	1.1715	0.7384	0.0000	0.6667	1.0955	1.6154	3.0000
APPG_FORM	0.0000	0.0000	3800.0000	1.3680	0.7620	0.0000	0.8000	1.4000	2.0000	3.0000
APPGA_FORM	0.0000	0.0000	3800.0000	1.1519	0.8027	0.0000	0.6000	1.0000	1.6000	3.0000
HGS	0.0000	0.0000	3800.0000	1.3713	0.6222	0.0000	1.0000	1.2857	1.7582	5.0000
HGSH	0.0000	0.0000	3800.0000	1.4520	0.7941	0.0000	1.0000	1.3798	1.8750	6.0000
HGS_FORM	0.0000	0.0000	3800.0000	1.3577	0.7377	0.0000	0.8000	1.2000	1.8000	5.0000
HGSH_FORM	0.0000	0.0000	3800.0000	1.4564	0.8534	0.0000	1.0000	1.4000	2.0000	6.0000
AGS	0.0000	0.0000	3800.0000	1.3796	0.6320	0.0000	1.0000	1.3000	1.7500	5.0000
AGSA	0.0000	0.0000	3800.0000	1.2261	0.6771	0.0000	0.8182	1.1538	1.6278	6.0000
AGS_FORM	0.0000	0.0000	3800.0000	1.3850	0.7415	0.0000	0.8000	1.4000	1.8000	5.0000
AGSA_FORM	0.0000	0.0000	3800.0000	1.2138	0.7459	0.0000	0.6667	1.2000	1.6000	6.0000
HGC	0.0000	0.0000	3800.0000	1.3774	0.5537	0.0000	1.0714	1.3750	1.7000	5.0000
HGCH	0.0000	0.0000	3800.0000	1.2241	0.6297	0.0000	0.8750	1.2143	1.5455	6.0000
HGC_FORM	0.0000	0.0000	3800.0000	1.3830	0.7042	0.0000	1.0000	1.4000	1.8000	5.0000
HGCH_FORM	0.0000	0.0000	3800.0000	1.2141	0.7189	0.0000	0.8000	1.2000	1.6000	6.0000
AGC	0.0000	0.0000	3800.0000	1.3735	0.5438	0.0000	1.0625	1.3636	1.6923	5.0000
AGCA	0.0000	0.0000	3800.0000	1.4484	0.7251	0.0000	1.0000	1.4615	1.9023	6.5000
AGC_FORM	0.0000	0.0000	3800.0000	1.3581	0.6901	0.0000	1.0000	1.3333	1.8000	5.0000
AGCA_FORM	0.0000	0.0000	3800.0000	1.4518	0.7982	0.0000	1.0000	1.4000	2.0000	6.5000
HCS	0.0000	0.0000	3800.0000	0.2619	0.1714	0.0000	0.1667	0.2500	0.3438	1.0000
HCSH	0.0000	0.0000	3800.0000	0.2771	0.2118	0.0000	0.1250	0.2727	0.3889	1.0000
HCS_FORM	0.0000	0.0000	3800.0000	0.2619	0.2292	0.0000	0.0000	0.2000	0.4000	1.0000
HCSH_FORM	0.0000	0.0000	3800.0000	0.2865	0.2471	0.0000	0.0000	0.2000	0.4000	1.0000

Numerical Feature Summary Statistics (cont.)

	missing	missing_%	count	mean	std	min	25%	median	75%	max
ACS	0.0000	0.0000	3800.0000	0.2613	0.1674	0.0000	0.1667	0.2500	0.3448	1.0000
ACSA	0.0000	0.0000	3800.0000	0.2326	0.2076	0.0000	0.0714	0.2000	0.3333	1.0000
ACS_FORM	0.0000	0.0000	3800.0000	0.2676	0.2287	0.0000	0.0000	0.2000	0.4000	1.0000
ACSA_FORM	0.0000	0.0000	3800.0000	0.2277	0.2379	0.0000	0.0000	0.2000	0.4000	1.0000
HPOS	0.0000	0.0000	3800.0000	10.7689	5.7188	1.0000	6.0000	11.0000	16.0000	20.0000
APOS	0.0000	0.0000	3800.0000	10.6545	5.6802	1.0000	6.0000	11.0000	15.0000	20.0000
HPDU	0.0000	0.0000	3800.0000	1.8534	2.6280	0.0000	0.0000	1.0000	2.0000	25.0000
HPDD	0.0000	0.0000	3800.0000	1.7455	2.5890	0.0000	0.0000	1.0000	2.0000	23.0000
APDU	0.0000	0.0000	3800.0000	1.8784	2.6698	0.0000	0.0000	1.0000	2.0000	25.0000
APDD	0.0000	0.0000	3800.0000	1.7734	2.6025	0.0000	0.0000	1.0000	2.0000	23.0000
HS_FORM	0.0000	0.0000	3800.0000	12.2102	3.7943	0.0000	10.0000	12.0000	14.4000	29.0000
AS_FORM	0.0000	0.0000	3800.0000	12.4605	3.8624	0.0000	10.2000	12.4000	14.6000	31.0000
HST_FORM	0.0000	0.0000	3800.0000	4.2232	1.6016	0.0000	3.2000	4.2000	5.2000	10.4000
AST_FORM	0.0000	0.0000	3800.0000	4.3110	1.6334	0.0000	3.2000	4.2000	5.2000	13.0000
HF_FORM	0.0000	0.0000	3800.0000	10.5198	2.5772	0.0000	9.4000	10.8000	12.0000	20.0000
AF_FORM	0.0000	0.0000	3800.0000	10.4849	2.5591	0.0000	9.4000	10.6000	12.0000	19.0000
HC_FORM	0.0000	0.0000	3800.0000	5.0316	1.7854	0.0000	4.0000	5.0000	6.0000	16.0000
AC_FORM	0.0000	0.0000	3800.0000	5.1452	1.8274	0.0000	4.0000	5.0000	6.2000	14.0000
HY_FORM	0.0000	0.0000	3800.0000	1.6999	0.7260	0.0000	1.2000	1.6000	2.2000	5.0000
AY_FORM	0.0000	0.0000	3800.0000	1.6783	0.7192	0.0000	1.2000	1.6000	2.2000	5.0000
HR_FORM	0.0000	0.0000	3800.0000	0.0593	0.1173	0.0000	0.0000	0.0000	0.0000	1.0000
AR_FORM	0.0000	0.0000	3800.0000	0.0593	0.1214	0.0000	0.0000	0.0000	0.0000	2.0000

HomeTeam (top 20 of 34)

	count	percentage
Chelsea	190.0000	5.0000
Everton	190.0000	5.0000
Man United	190.0000	5.0000
Arsenal	190.0000	5.0000
Tottenham	190.0000	5.0000
West Ham	190.0000	5.0000
Crystal Palace	190.0000	5.0000
Man City	190.0000	5.0000
Liverpool	190.0000	5.0000
Leicester	171.0000	4.5000
Newcastle	171.0000	4.5000
Southampton	171.0000	4.5000
Bournemouth	152.0000	4.0000
Brighton	152.0000	4.0000
Aston Villa	133.0000	3.5000
Burnley	133.0000	3.5000
Wolves	133.0000	3.5000
Watford	114.0000	3.0000
Fulham	95.0000	2.5000
West Brom	76.0000	2.0000

Referee (top 20 of 49)

	count	percentage
A Taylor	295.0000	7.7600
M Oliver	284.0000	7.4700
C Pawson	236.0000	6.2100
M Atkinson	197.0000	5.1800
M Dean	195.0000	5.1300
A Marriner	188.0000	4.9500
J Moss	185.0000	4.8700
P Tierney	182.0000	4.7900
S Attwell	174.0000	4.5800
C Kavanagh	163.0000	4.2900
K Friend	156.0000	4.1100
S Hooper	125.0000	3.2900
D Coote	112.0000	2.9500
A Madley	112.0000	2.9500
L Mason	105.0000	2.7600
G Scott	104.0000	2.7400
P Bankes	93.0000	2.4500
R Jones	92.0000	2.4200
R Madley	76.0000	2.0000
D England	76.0000	2.0000

Categorical Feature Value Counts

AwayTeam (top 20 of 34)

	count	percentage
Tottenham	190.0000	5.0000
Crystal Palace	190.0000	5.0000
West Ham	190.0000	5.0000
Liverpool	190.0000	5.0000
Man City	190.0000	5.0000
Man United	190.0000	5.0000
Everton	190.0000	5.0000
Arsenal	190.0000	5.0000
Chelsea	190.0000	5.0000
Southampton	171.0000	4.5000
Newcastle	171.0000	4.5000
Leicester	171.0000	4.5000
Bournemouth	152.0000	4.0000
Brighton	152.0000	4.0000
Aston Villa	133.0000	3.5000
Burnley	133.0000	3.5000
Wolves	133.0000	3.5000
Watford	114.0000	3.0000
Fulham	95.0000	2.5000
West Brom	76.0000	2.0000

FTR (top 3 of 3)

	count	percentage
H	1691.0000	44.5000
A	1223.0000	32.1800
D	886.0000	23.3200

Categorical Summary — HomeTeam

	count	percentage
Chelsea	190.0000	5.0000
Everton	190.0000	5.0000
Man United	190.0000	5.0000
Arsenal	190.0000	5.0000
Tottenham	190.0000	5.0000
West Ham	190.0000	5.0000
Crystal Palace	190.0000	5.0000
Man City	190.0000	5.0000
Liverpool	190.0000	5.0000
Leicester	171.0000	4.5000
Newcastle	171.0000	4.5000
Southampton	171.0000	4.5000
Bournemouth	152.0000	4.0000
Brighton	152.0000	4.0000
Aston Villa	133.0000	3.5000
Burnley	133.0000	3.5000
Wolves	133.0000	3.5000
Watford	114.0000	3.0000
Fulham	95.0000	2.5000
West Brom	76.0000	2.0000
Brentford	76.0000	2.0000
Norwich	57.0000	1.5000
Stoke	57.0000	1.5000
Swansea	57.0000	1.5000
Sheffield United	57.0000	1.5000
Leeds	57.0000	1.5000
Nott'm Forest	57.0000	1.5000
Sunderland	38.0000	1.0000
Huddersfield	38.0000	1.0000
Hull	19.0000	0.5000

Categorical Summary — HomeTeam (cont.)

	count	percentage
Middlesbrough	19.0000	0.5000
Cardiff	19.0000	0.5000
Luton	19.0000	0.5000
Ipswich	19.0000	0.5000

Categorical Summary — AwayTeam

	count	percentage
Tottenham	190.0000	5.0000
Crystal Palace	190.0000	5.0000
West Ham	190.0000	5.0000
Liverpool	190.0000	5.0000
Man City	190.0000	5.0000
Man United	190.0000	5.0000
Everton	190.0000	5.0000
Arsenal	190.0000	5.0000
Chelsea	190.0000	5.0000
Southampton	171.0000	4.5000
Newcastle	171.0000	4.5000
Leicester	171.0000	4.5000
Bournemouth	152.0000	4.0000
Brighton	152.0000	4.0000
Aston Villa	133.0000	3.5000
Burnley	133.0000	3.5000
Wolves	133.0000	3.5000
Watford	114.0000	3.0000
Fulham	95.0000	2.5000
West Brom	76.0000	2.0000
Brentford	76.0000	2.0000
Swansea	57.0000	1.5000
Norwich	57.0000	1.5000
Stoke	57.0000	1.5000
Sheffield United	57.0000	1.5000
Leeds	57.0000	1.5000
Nott'm Forest	57.0000	1.5000
Sunderland	38.0000	1.0000
Huddersfield	38.0000	1.0000
Hull	19.0000	0.5000

Categorical Summary — AwayTeam (cont.)

	count	percentage
Middlesbrough	19.0000	0.5000
Cardiff	19.0000	0.5000
Luton	19.0000	0.5000
Ipswich	19.0000	0.5000

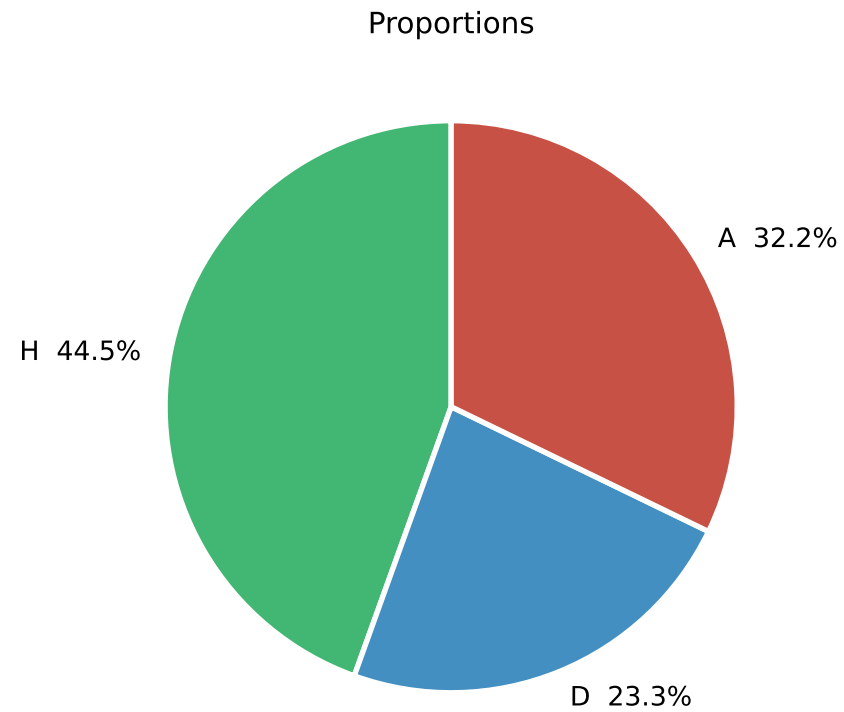
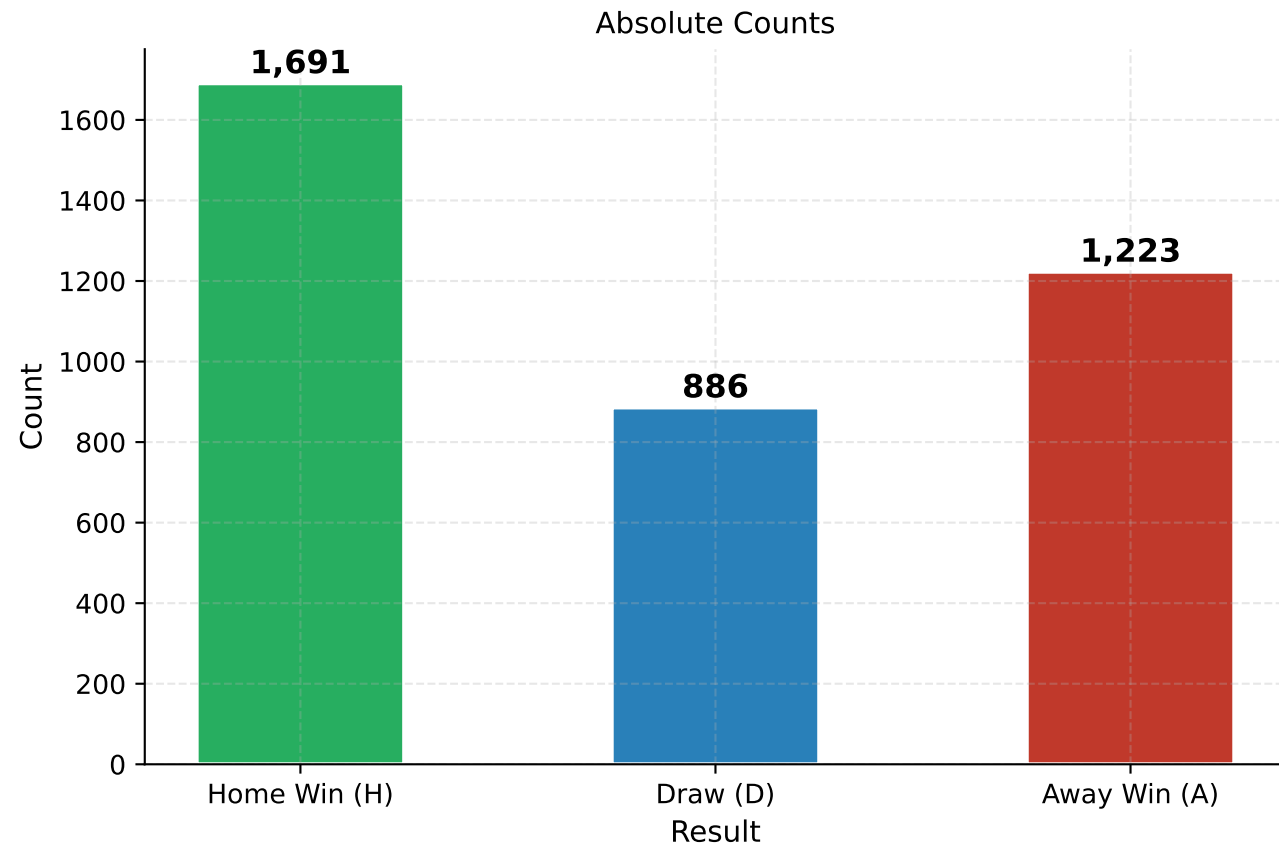
Categorical Summary — Referee

	count	percentage
A Taylor	295.0000	7.7600
M Oliver	284.0000	7.4700
C Pawson	236.0000	6.2100
M Atkinson	197.0000	5.1800
M Dean	195.0000	5.1300
A Marriner	188.0000	4.9500
J Moss	185.0000	4.8700
P Tierney	182.0000	4.7900
S Attwell	174.0000	4.5800
C Kavanagh	163.0000	4.2900
K Friend	156.0000	4.1100
S Hooper	125.0000	3.2900
D Coote	112.0000	2.9500
A Madley	112.0000	2.9500
L Mason	105.0000	2.7600
G Scott	104.0000	2.7400
P Bankes	93.0000	2.4500
R Jones	92.0000	2.4200
R Madley	76.0000	2.0000
D England	76.0000	2.0000
R East	64.0000	1.6800
N Swarbrick	62.0000	1.6300
J Gillett	62.0000	1.6300
J Brooks	62.0000	1.6300
M Jones	58.0000	1.5300
M Clattenburg	52.0000	1.3700
M Salisbury	47.0000	1.2400
T Robinson	40.0000	1.0500
S Barrott	38.0000	1.0000
T Harrington	37.0000	0.9700

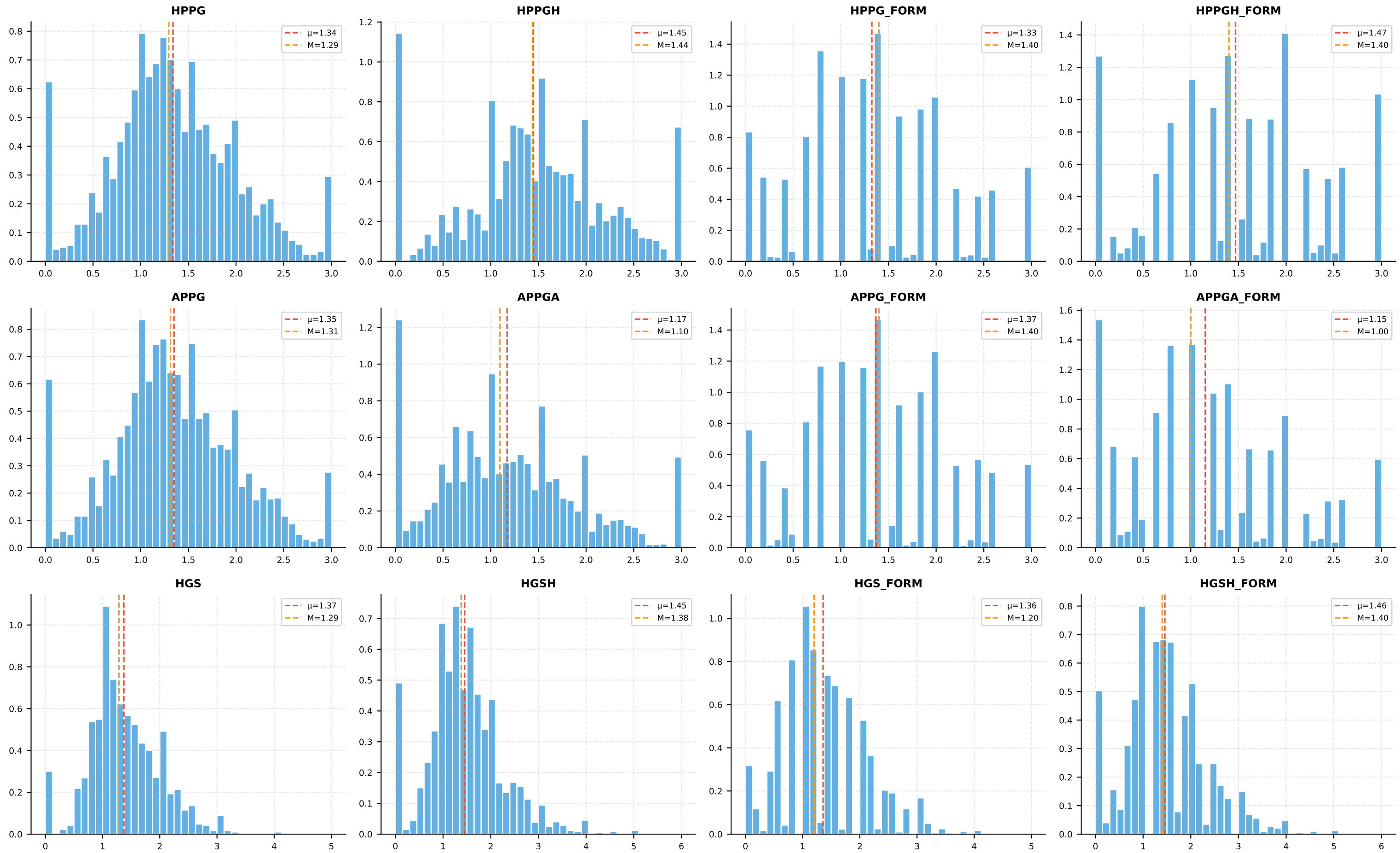
Categorical Summary — Referee (cont.)

	count	percentage
L Probert	36.0000	0.9500
T Bramall	30.0000	0.7900
D Bond	23.0000	0.6100
L Smith	9.0000	0.2400
J Smith	8.0000	0.2100
S Allison	4.0000	0.1100
K Stroud	2.0000	0.0500
R Welch	2.0000	0.0500
M Donohue	2.0000	0.0500
A Kitchen	2.0000	0.0500
D Webb	2.0000	0.0500
I Mason	1.0000	0.0300
O Langford	1.0000	0.0300
S Scott	1.0000	0.0300
A Moss	1.0000	0.0300
S Singh	1.0000	0.0300
J Gillett	1.0000	0.0300
K Kavanagh	1.0000	0.0300
O Oliver	1.0000	0.0300

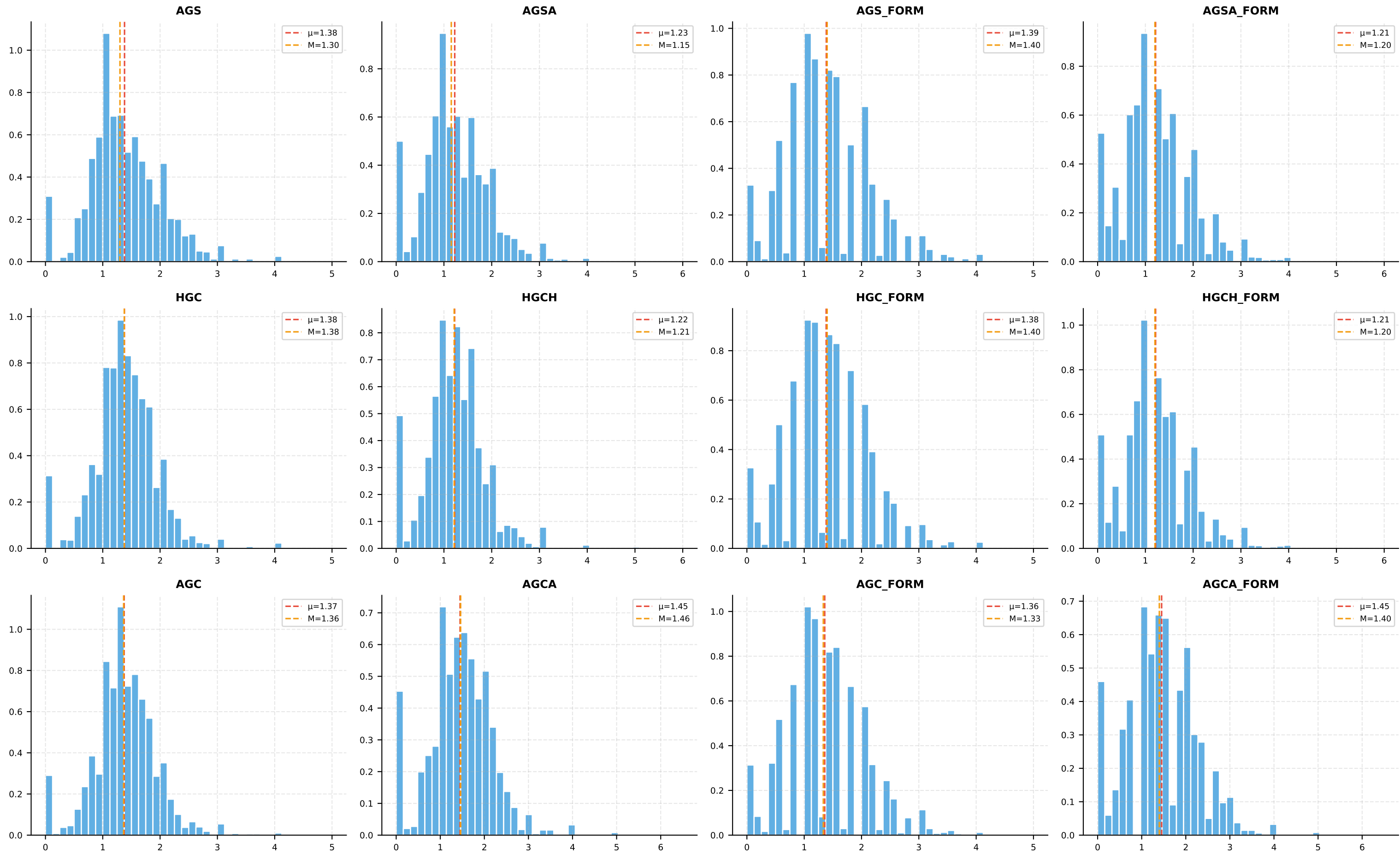
Target Variable: Full-Time Result (FTR)



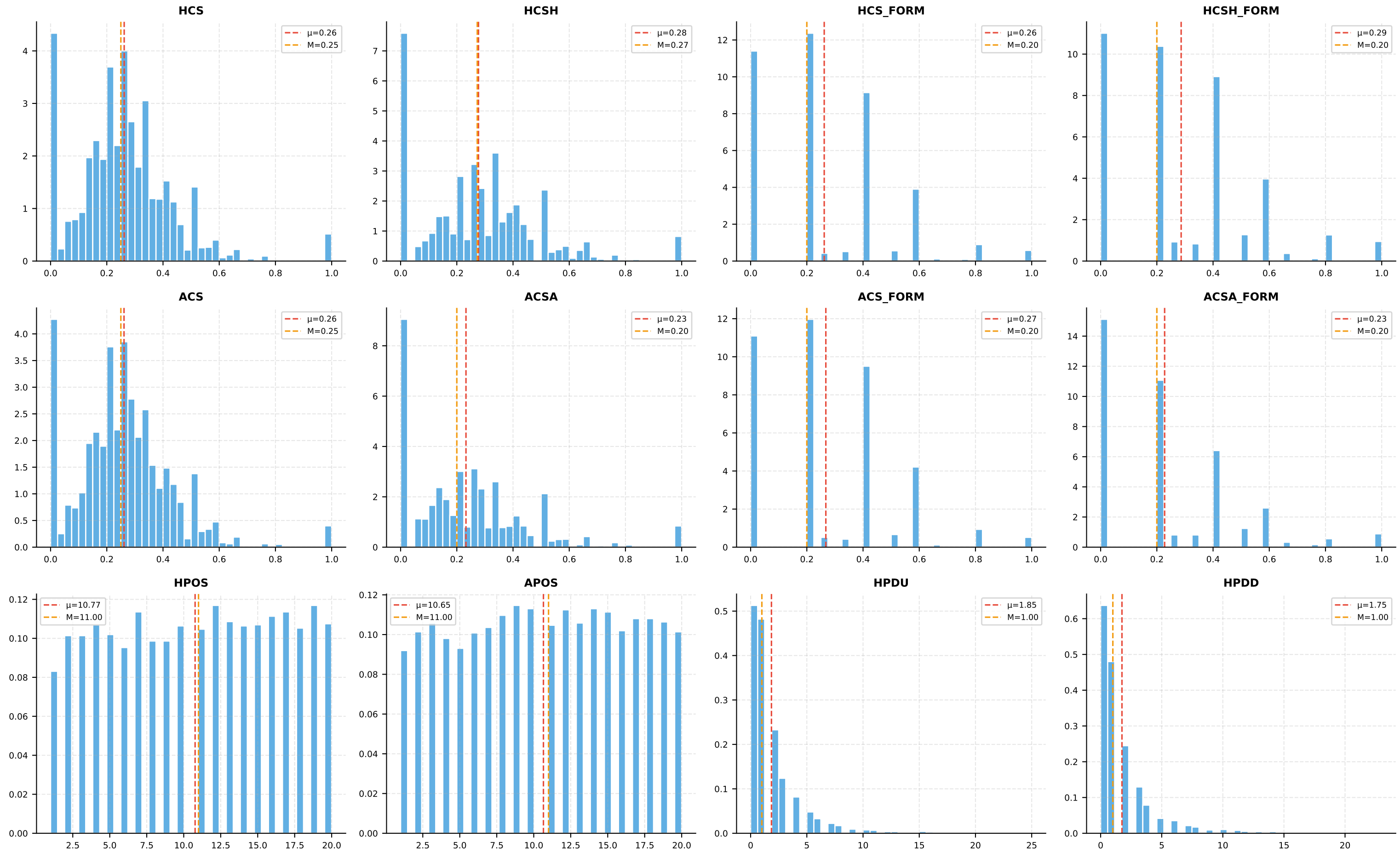
Numerical Feature Distributions



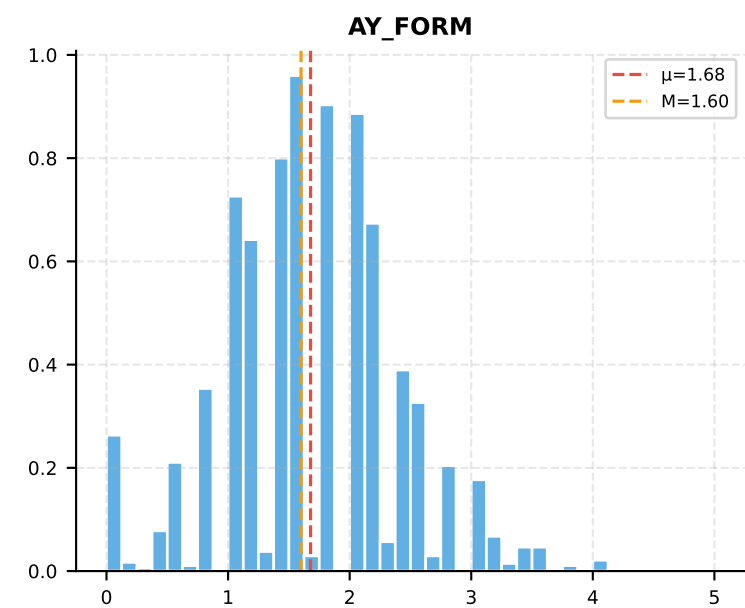
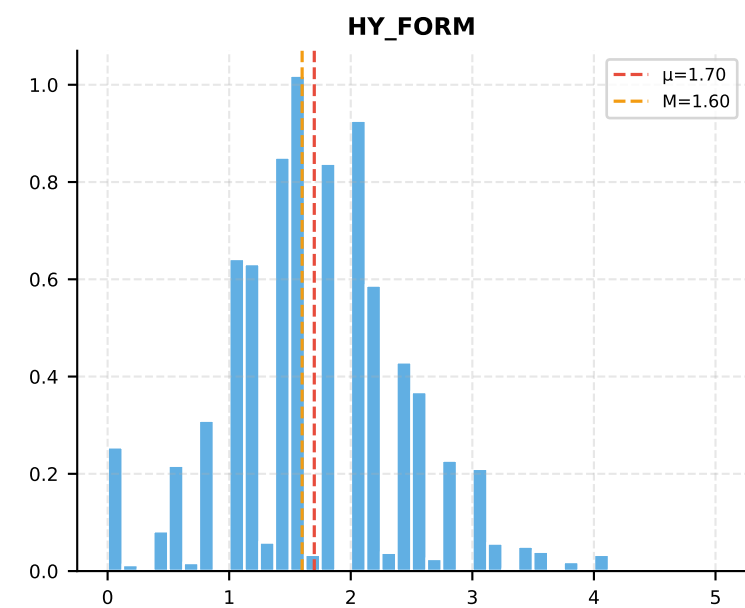
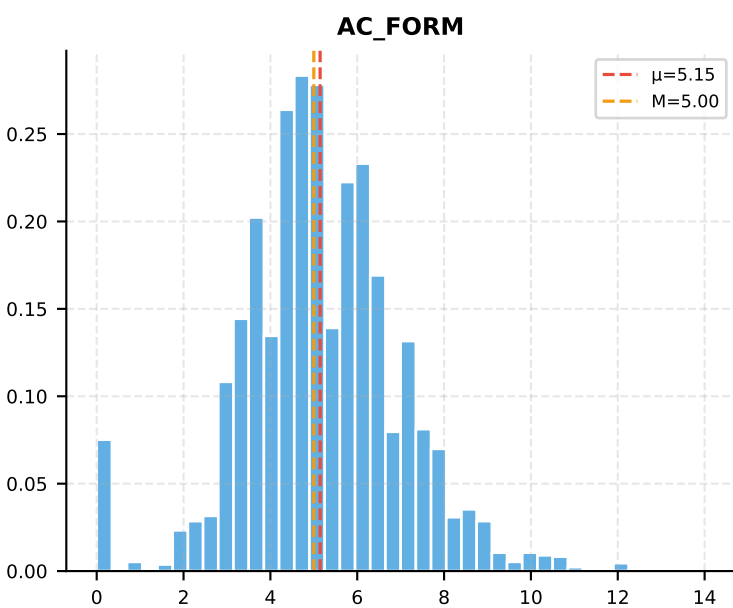
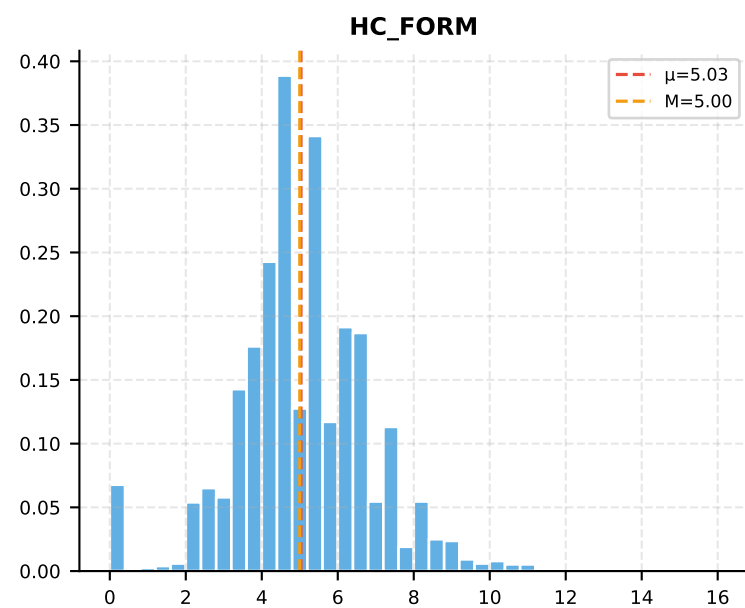
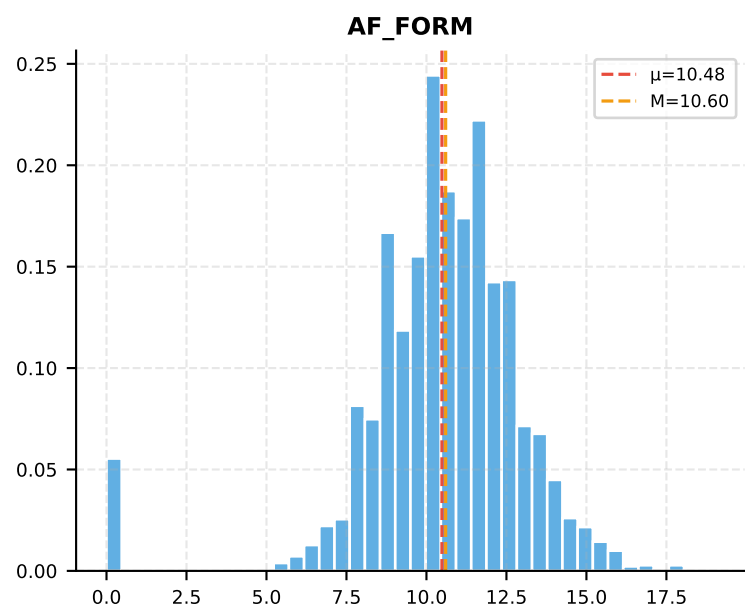
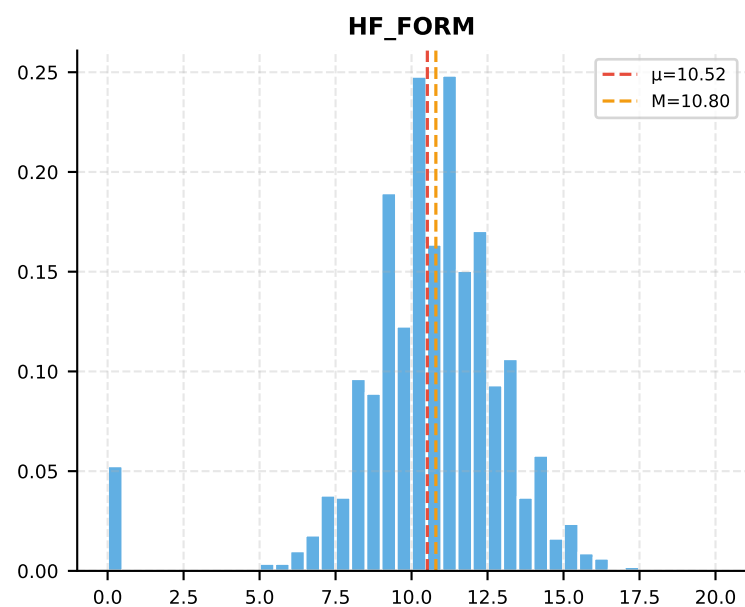
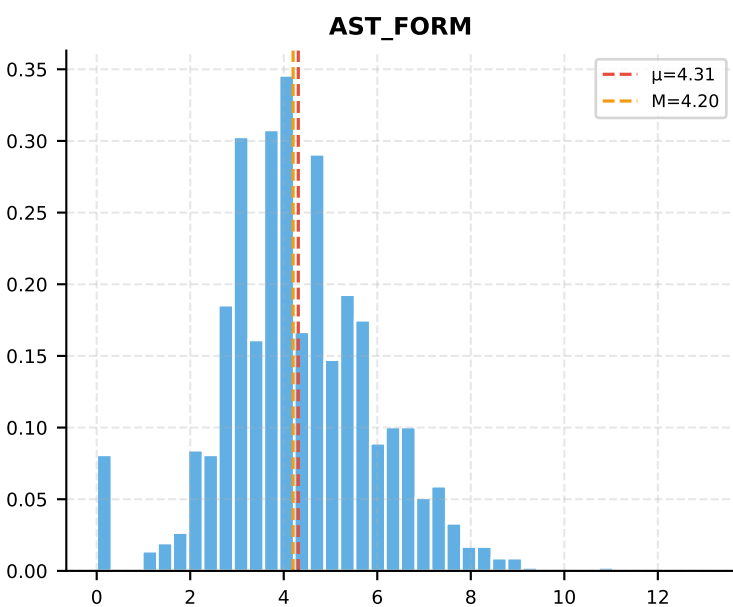
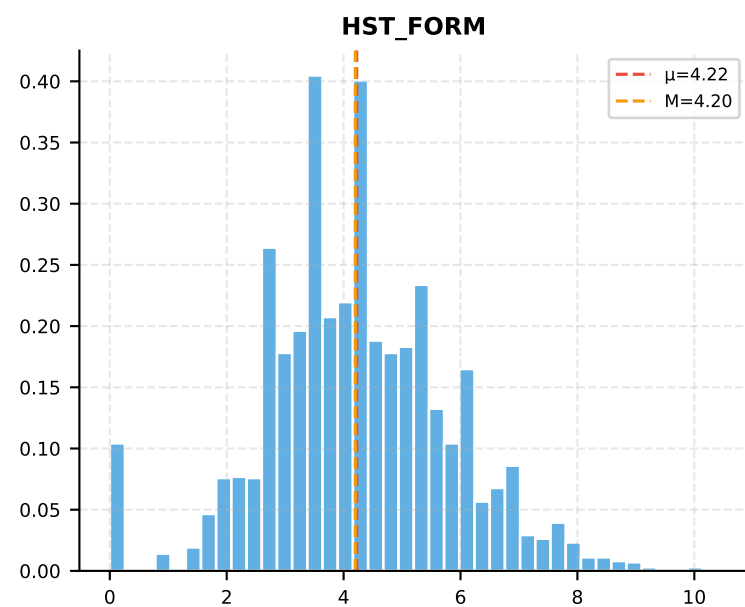
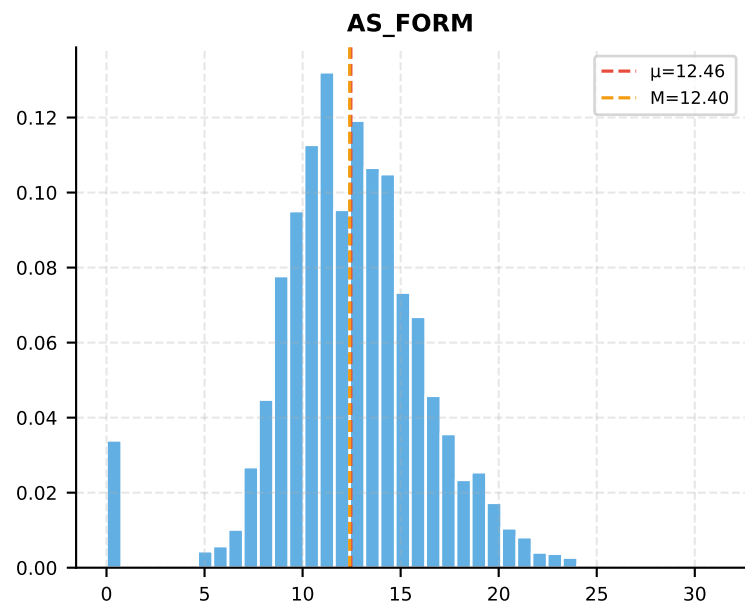
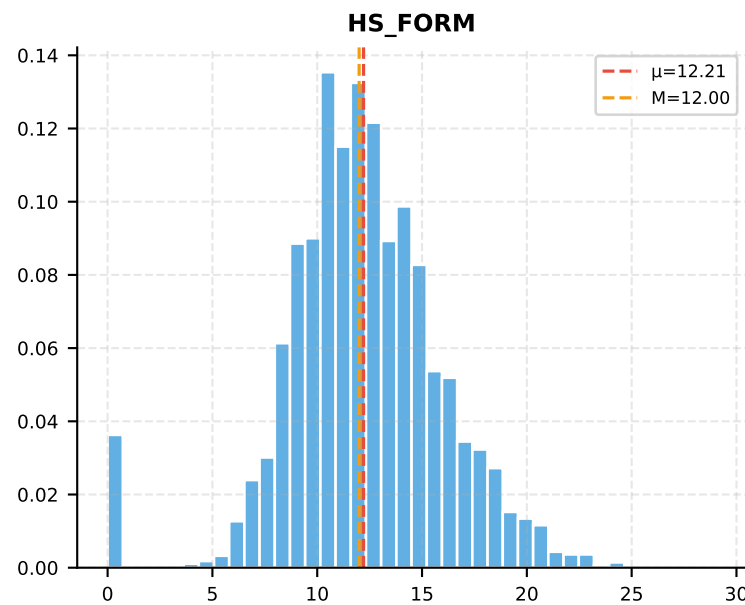
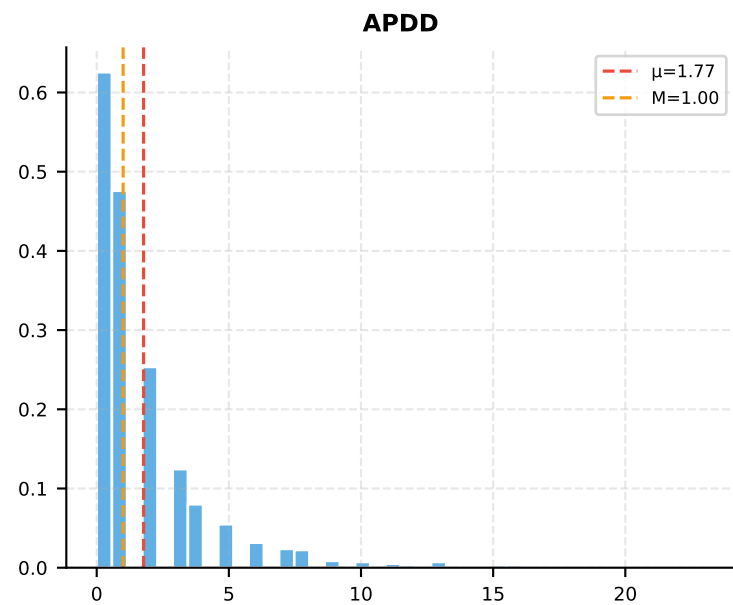
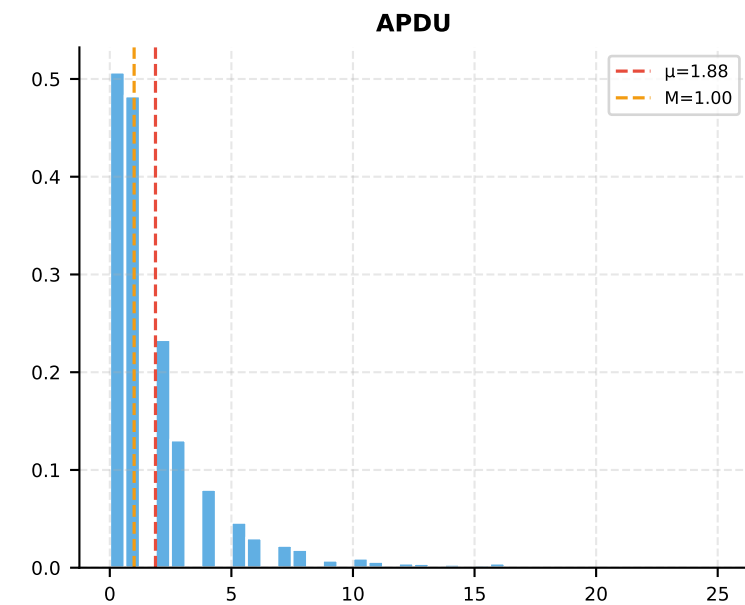
Numerical Feature Distributions



Numerical Feature Distributions

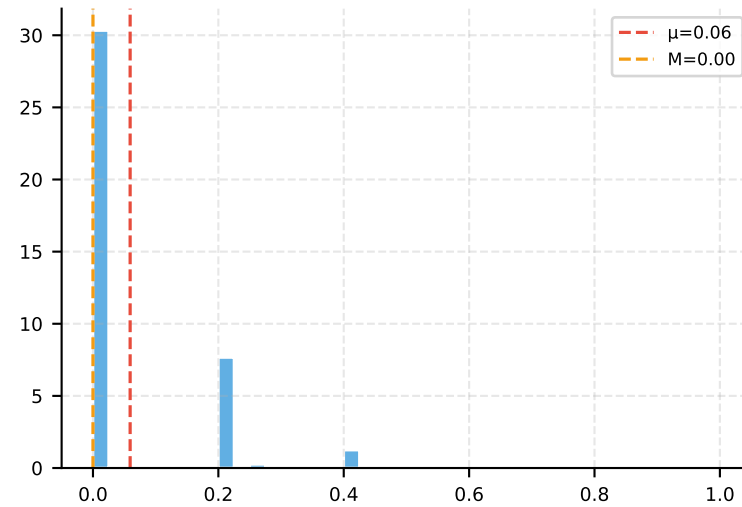


Numerical Feature Distributions

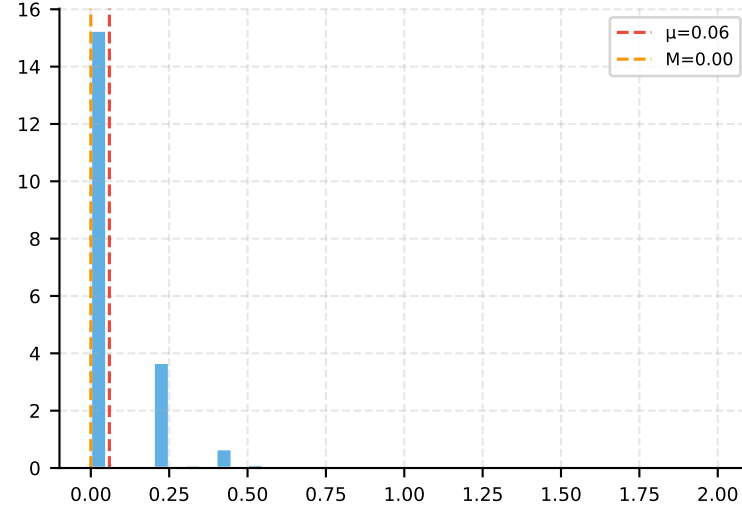


Numerical Feature Distributions

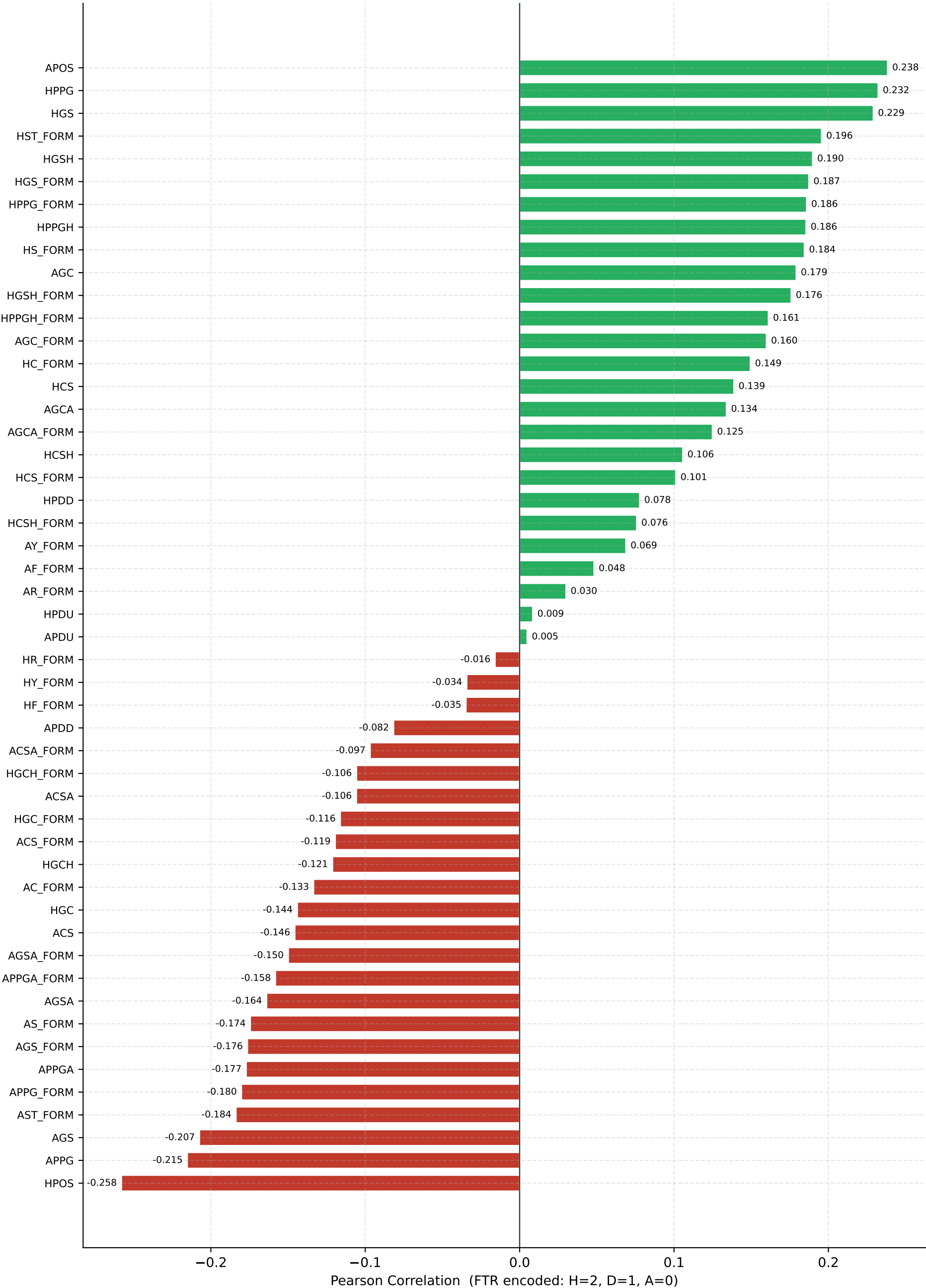
HR_FORM



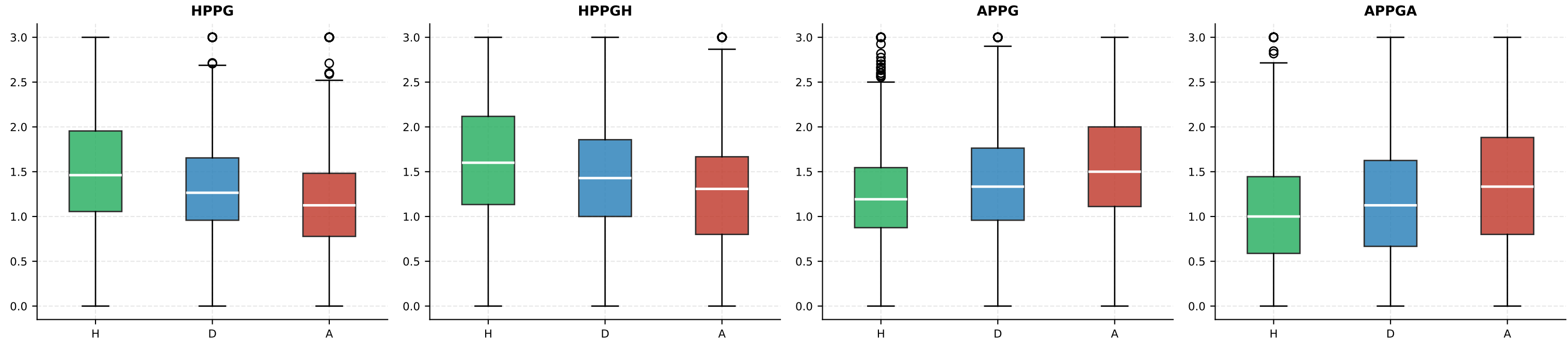
AR_FORM



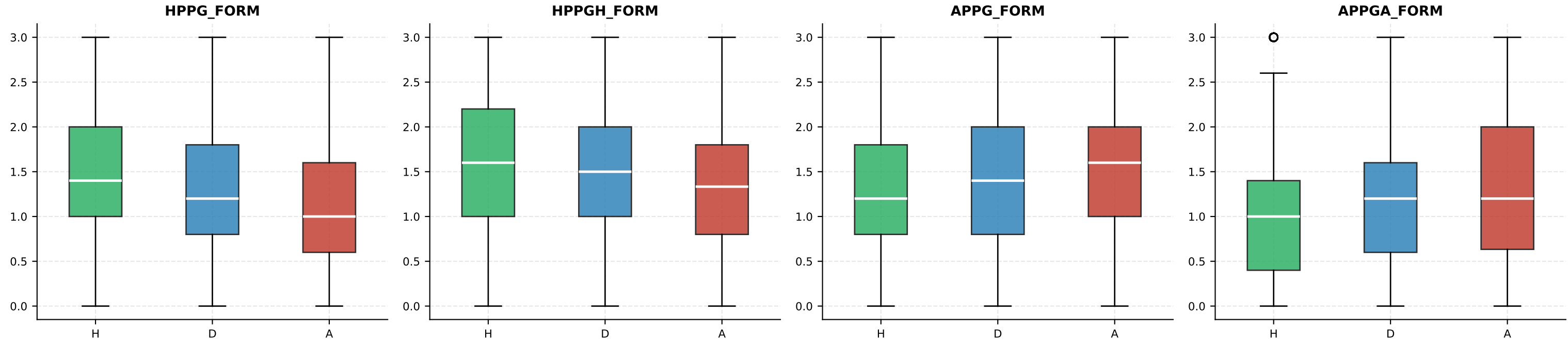
Feature Correlation with Target Variable (FTR)



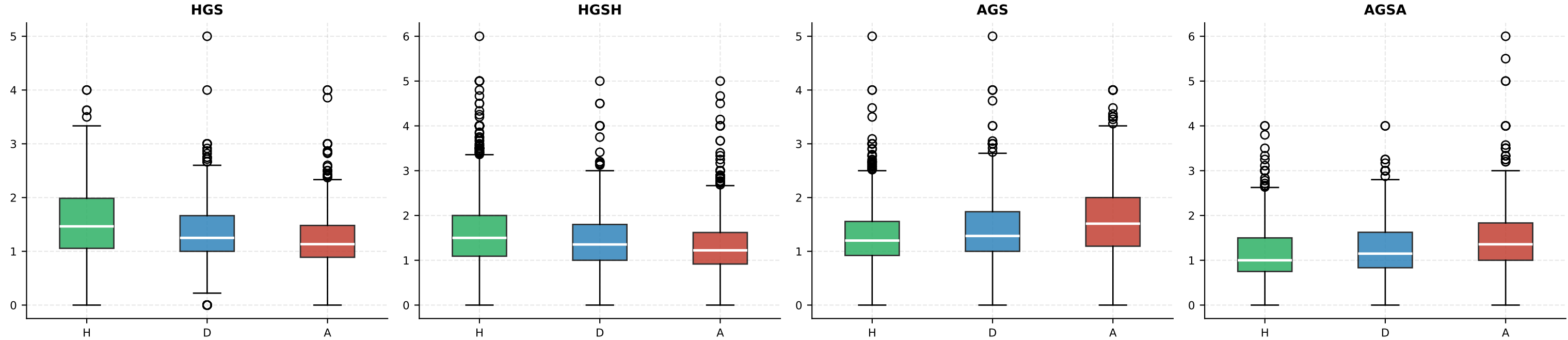
Distribution by FTR — Points Per Game (Season)



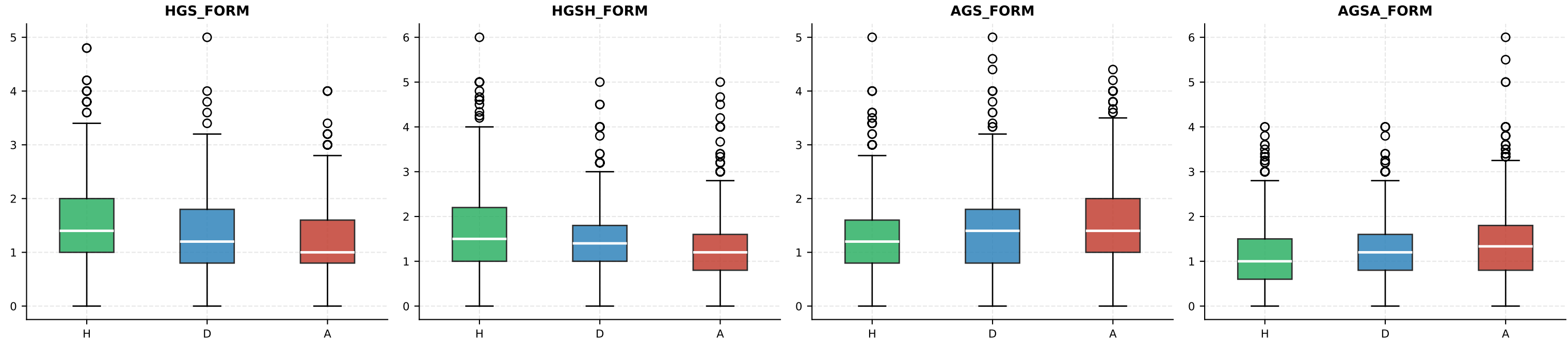
Distribution by FTR — Points Per Game (Form)



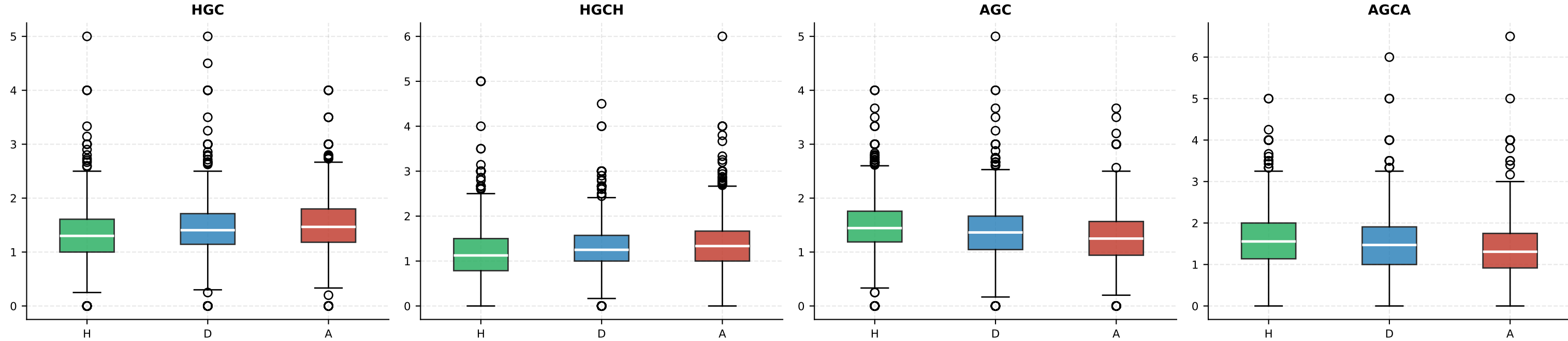
Distribution by FTR — Goals Scored (Season)



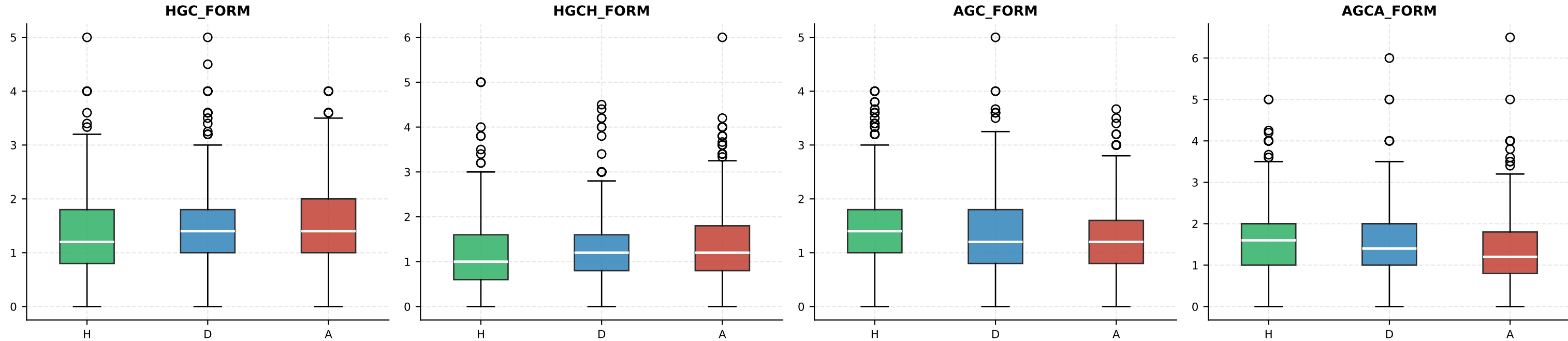
Distribution by FTR — Goals Scored (Form)



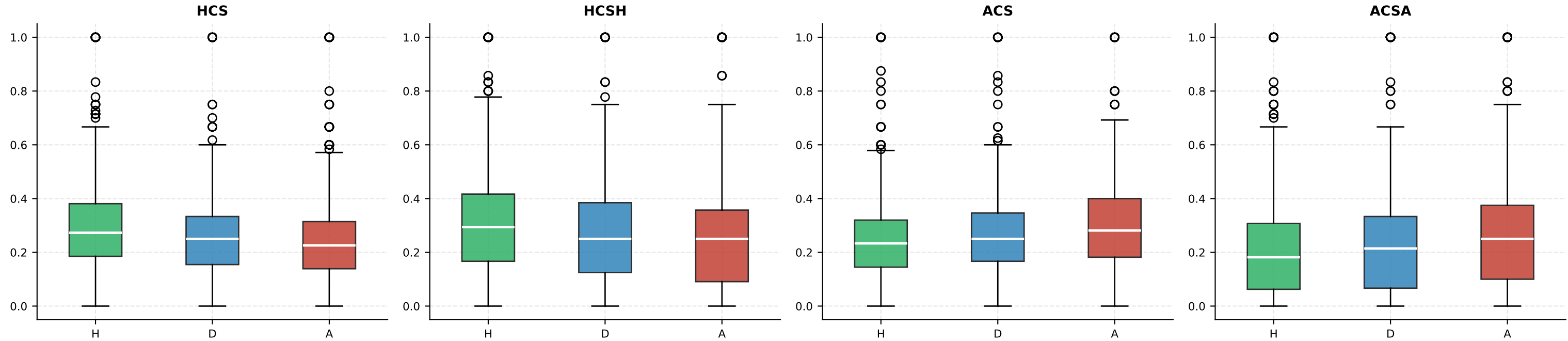
Distribution by FTR — Goals Conceded (Season)



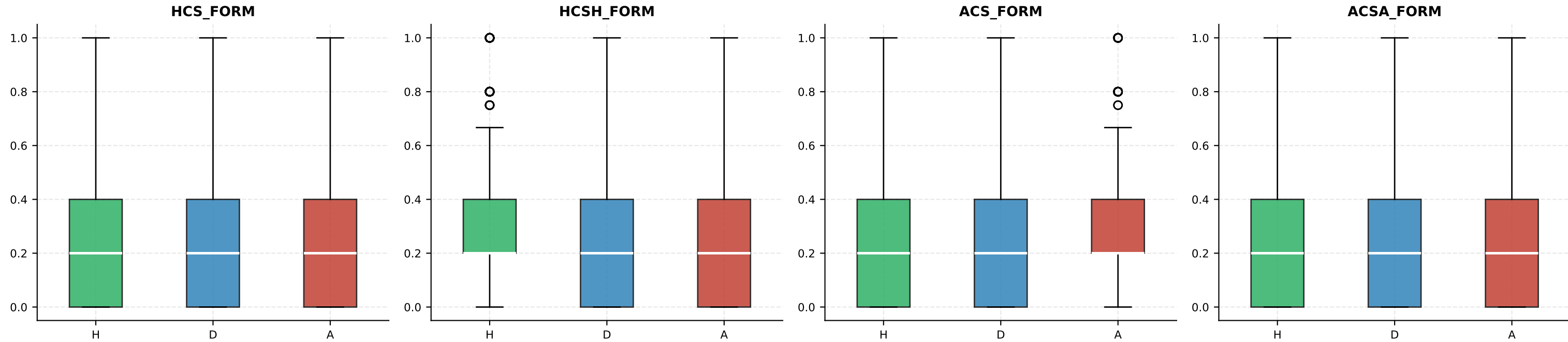
Distribution by FTR — Goals Conceded (Form)



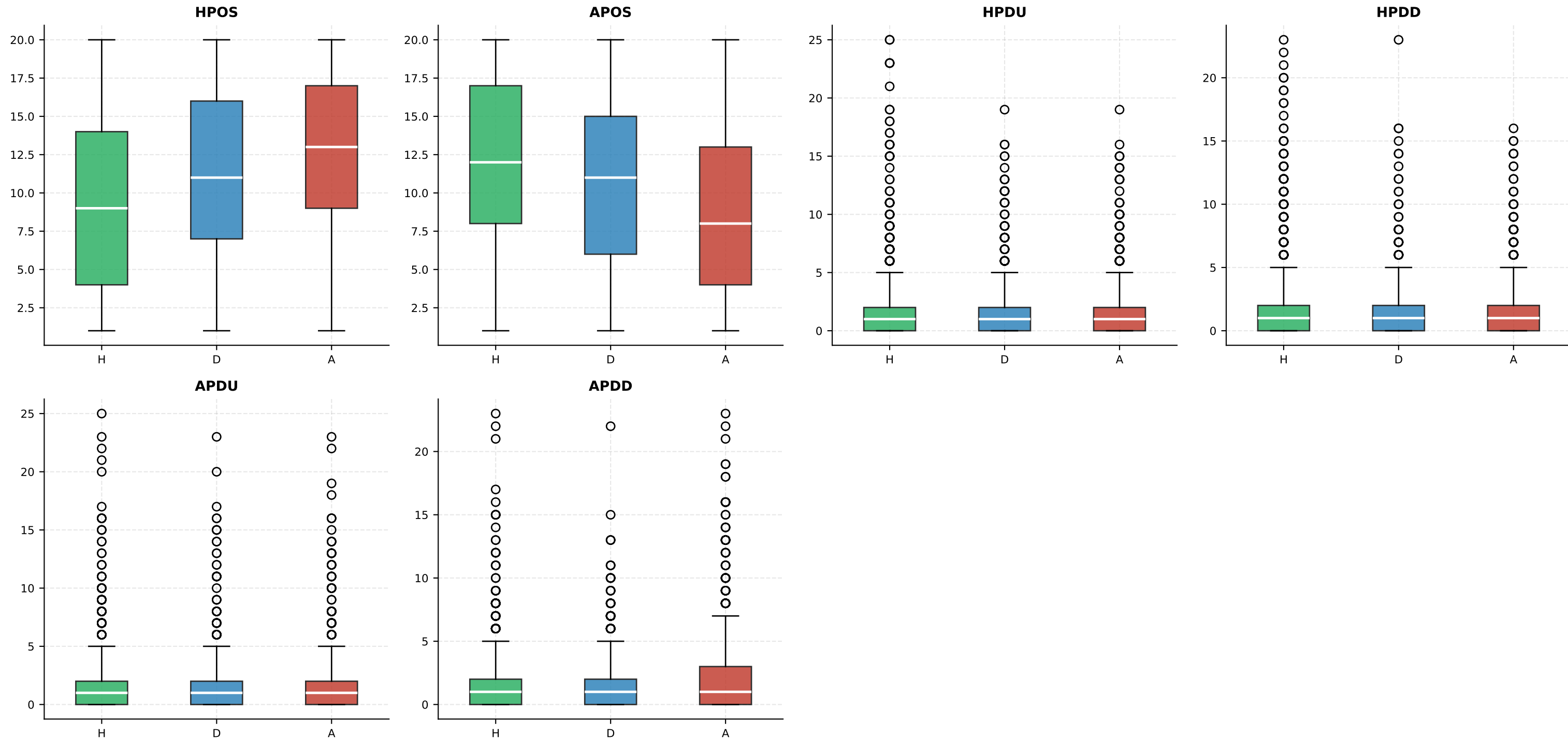
Distribution by FTR — Clean Sheets (Season)



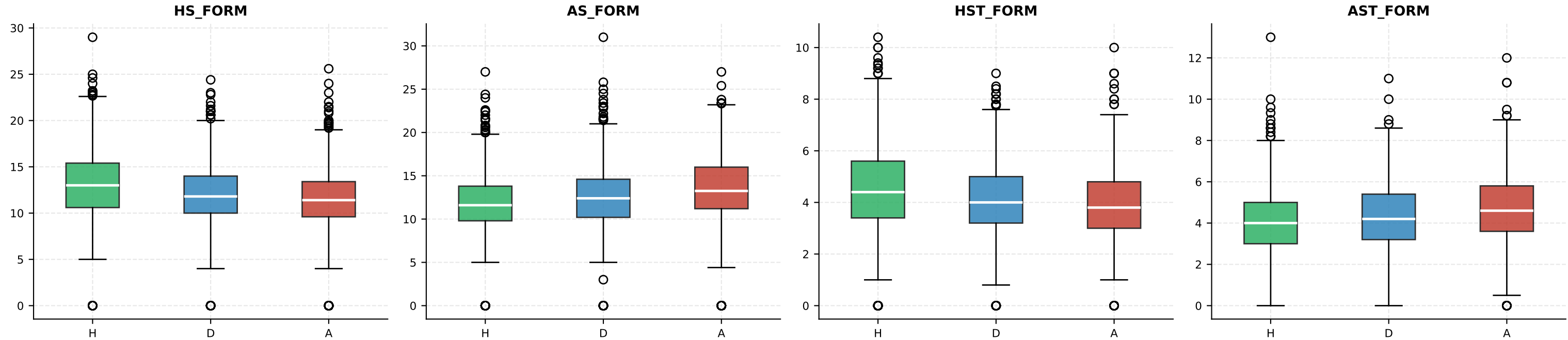
Distribution by FTR — Clean Sheets (Form)



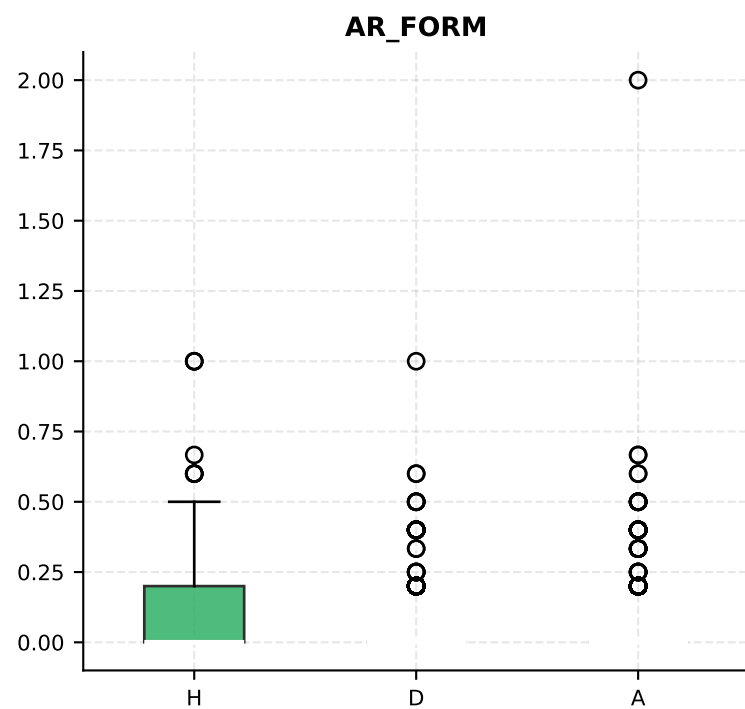
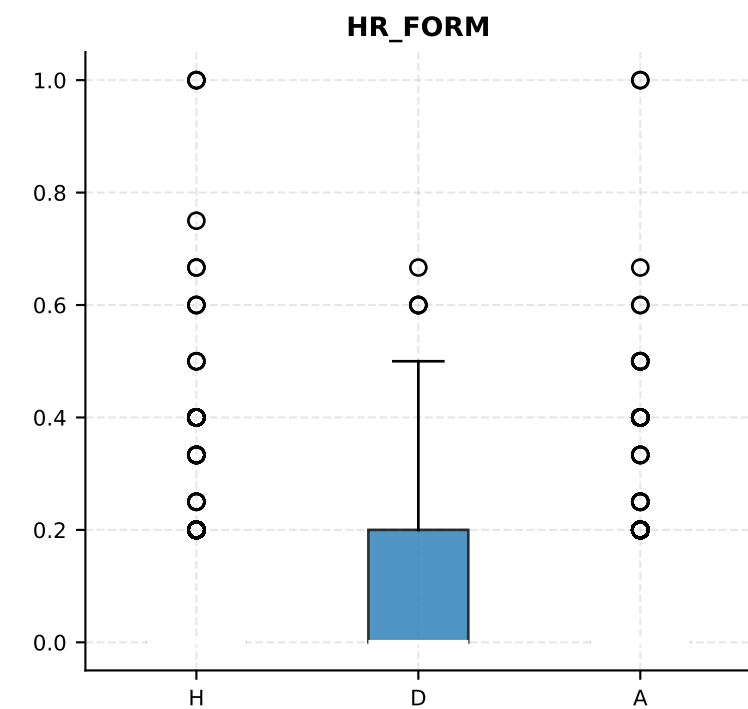
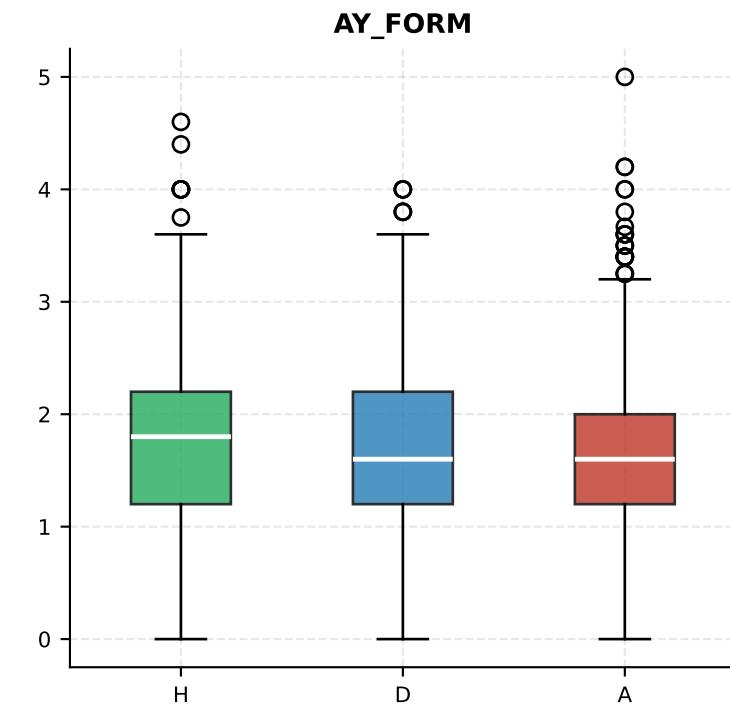
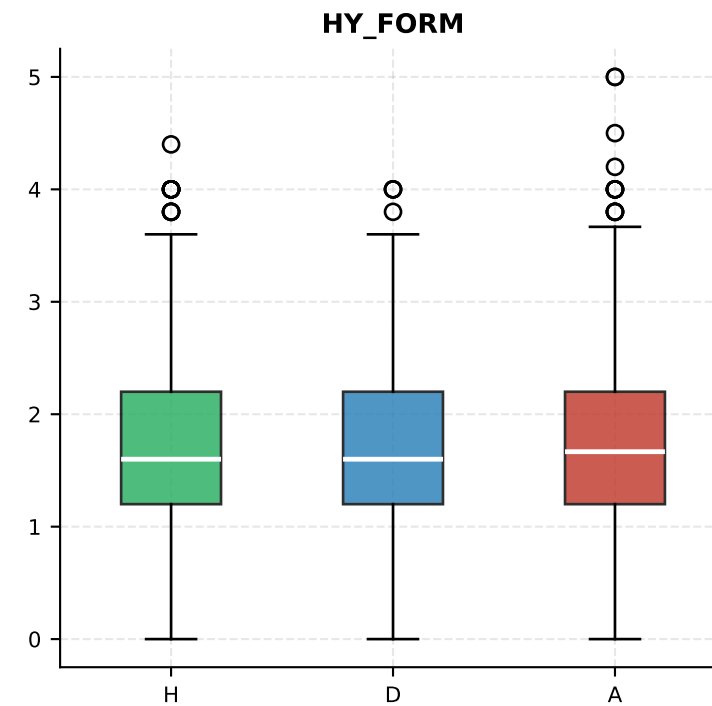
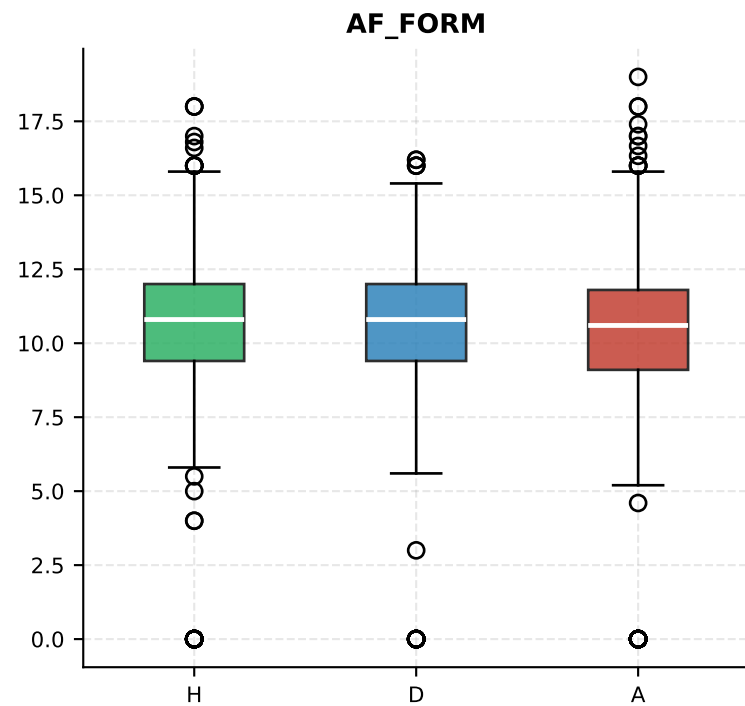
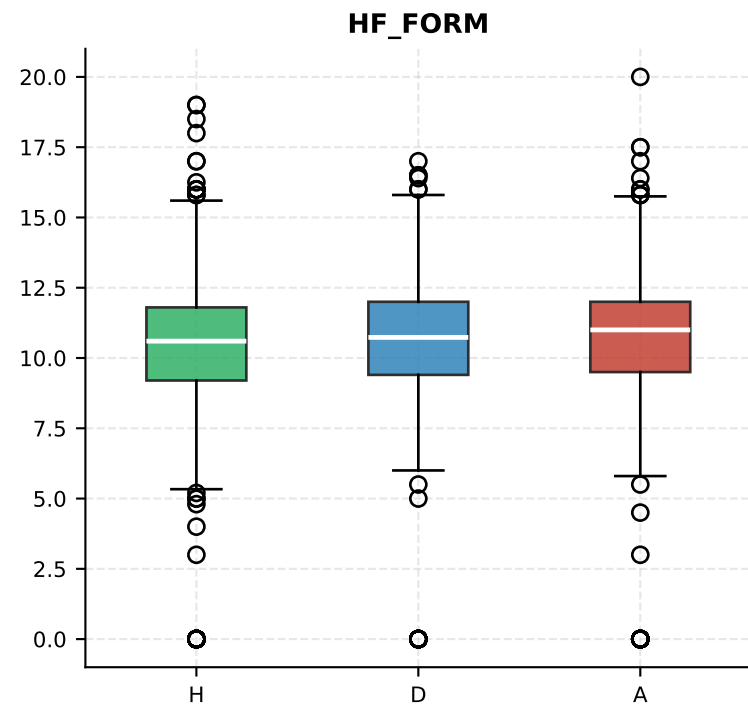
Distribution by FTR — League Position & Gap



Distribution by FTR — Shots (Form)



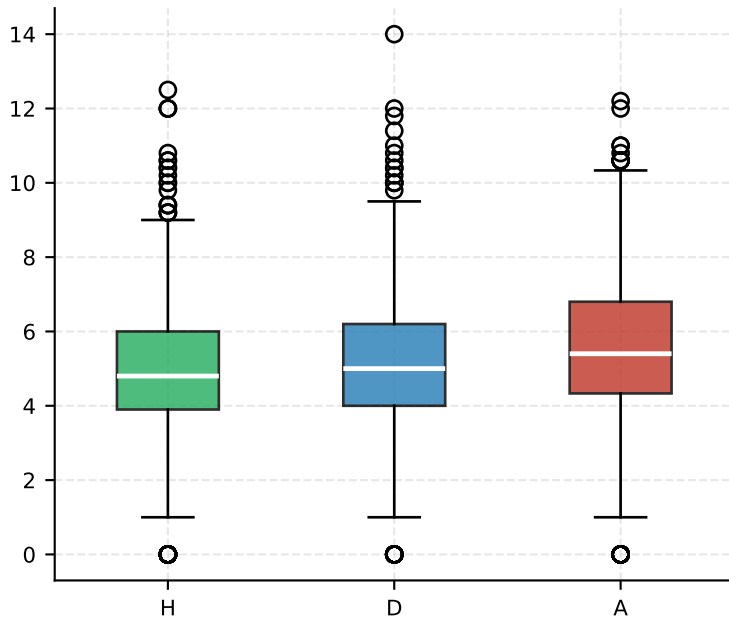
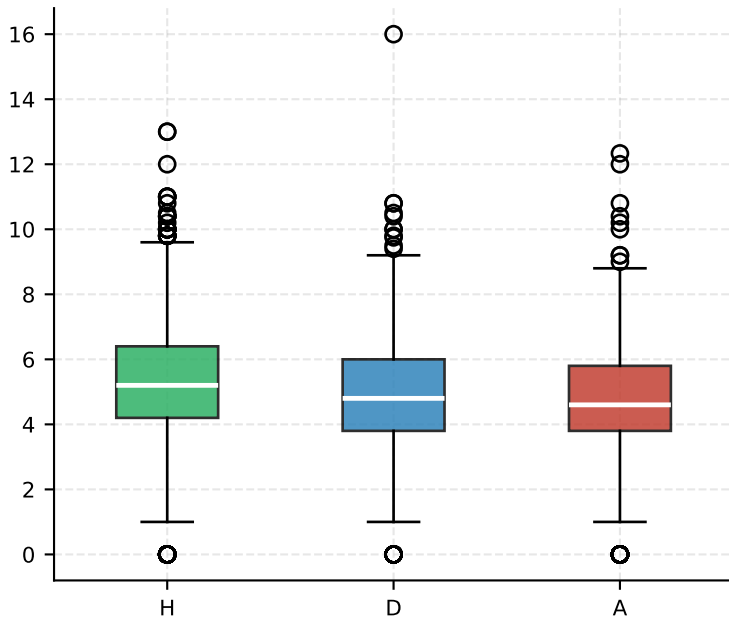
Distribution by FTR — Fouls & Cards (Form)



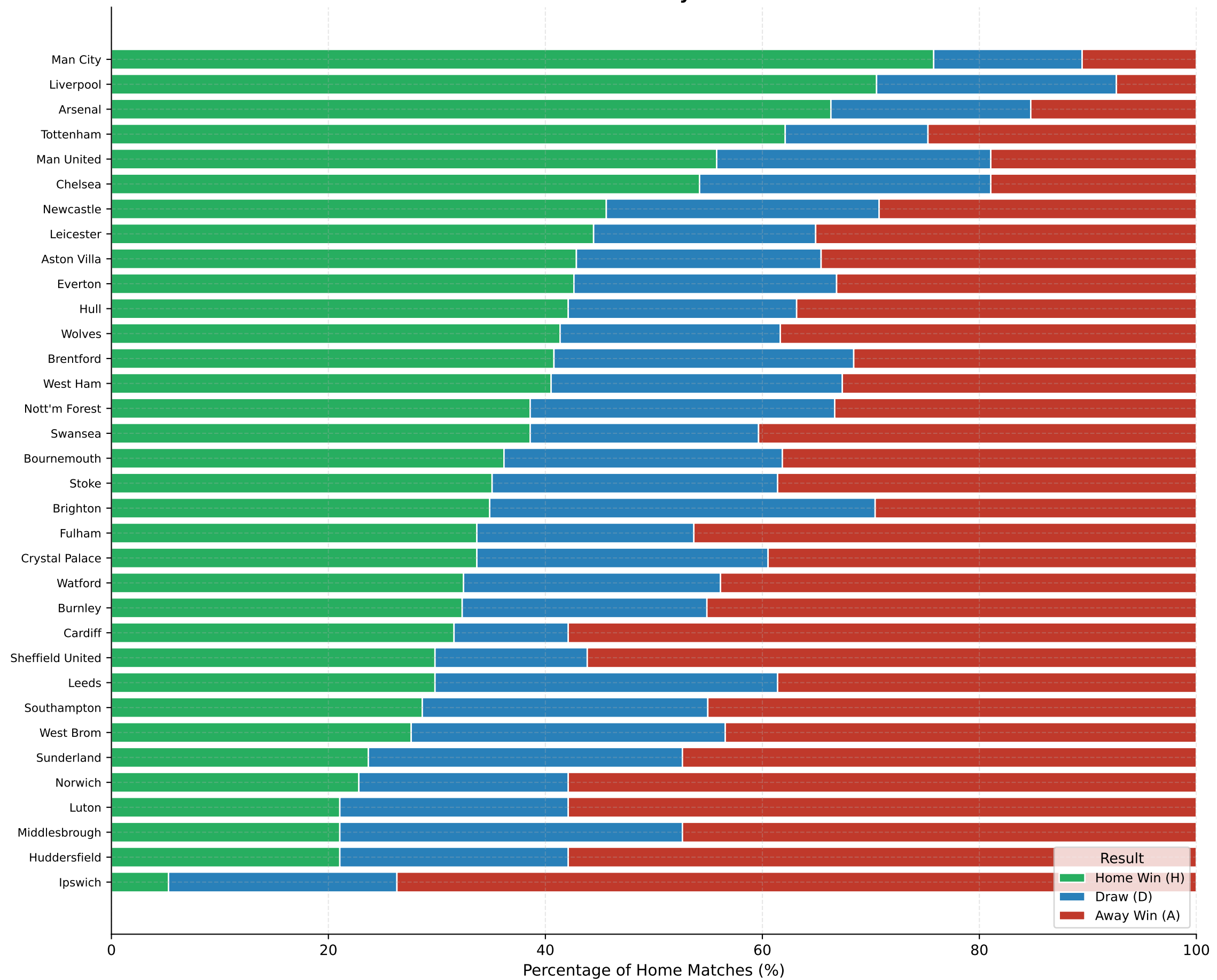
Distribution by FTR — Corners (Form)

HC_FORM

AC_FORM

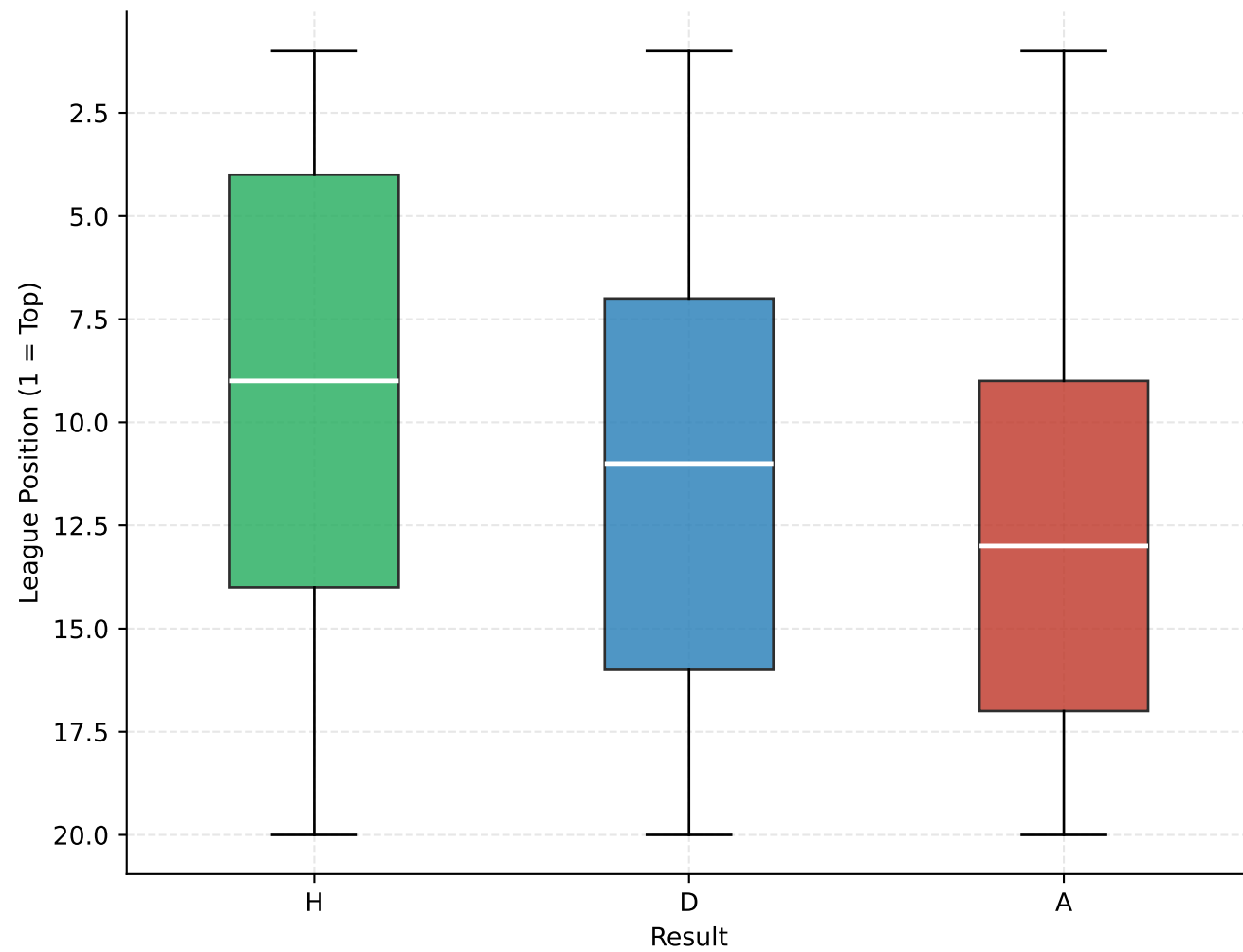


FTR Distribution by Home Team

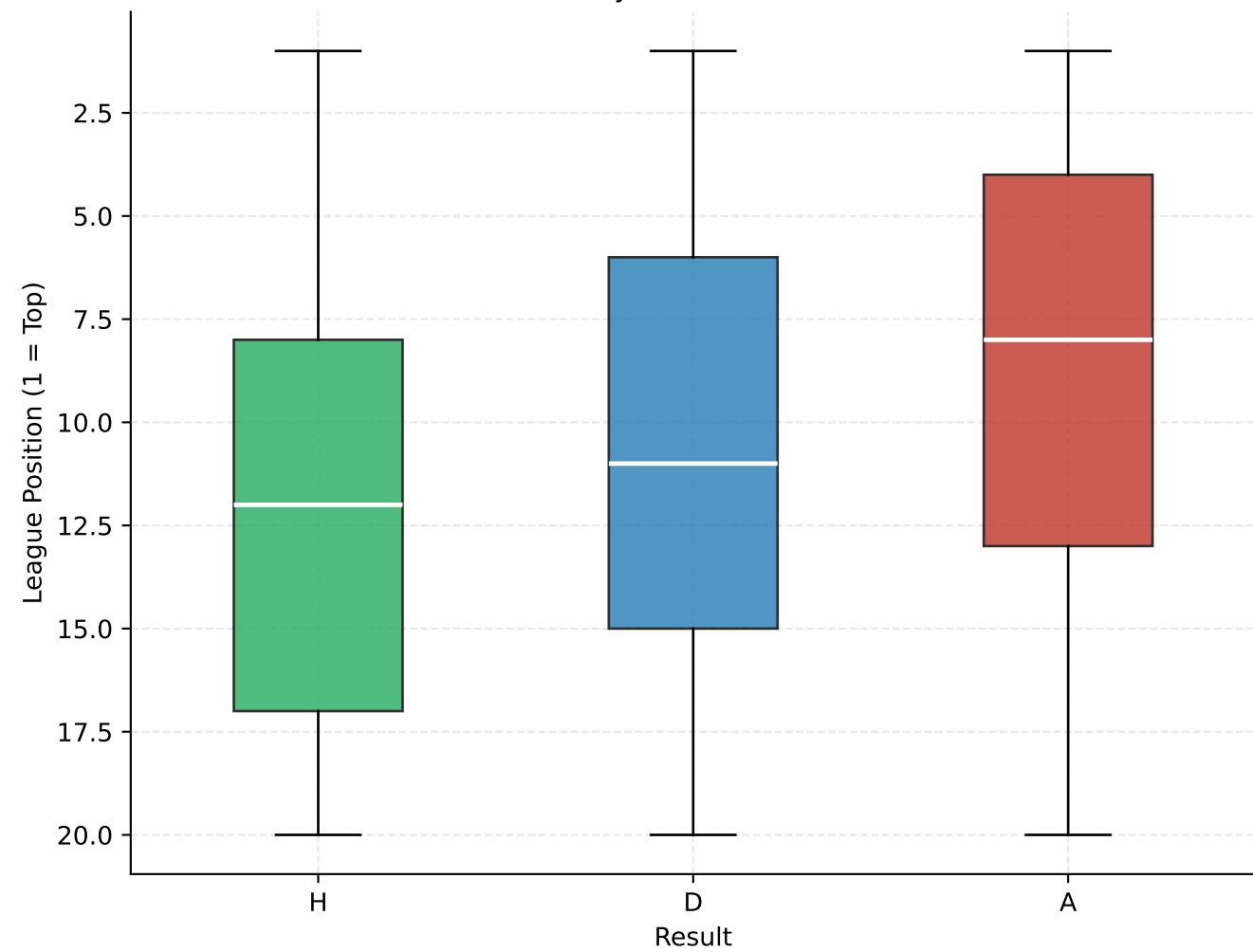


League Position at Kick-off vs Match Result

Home Team Position

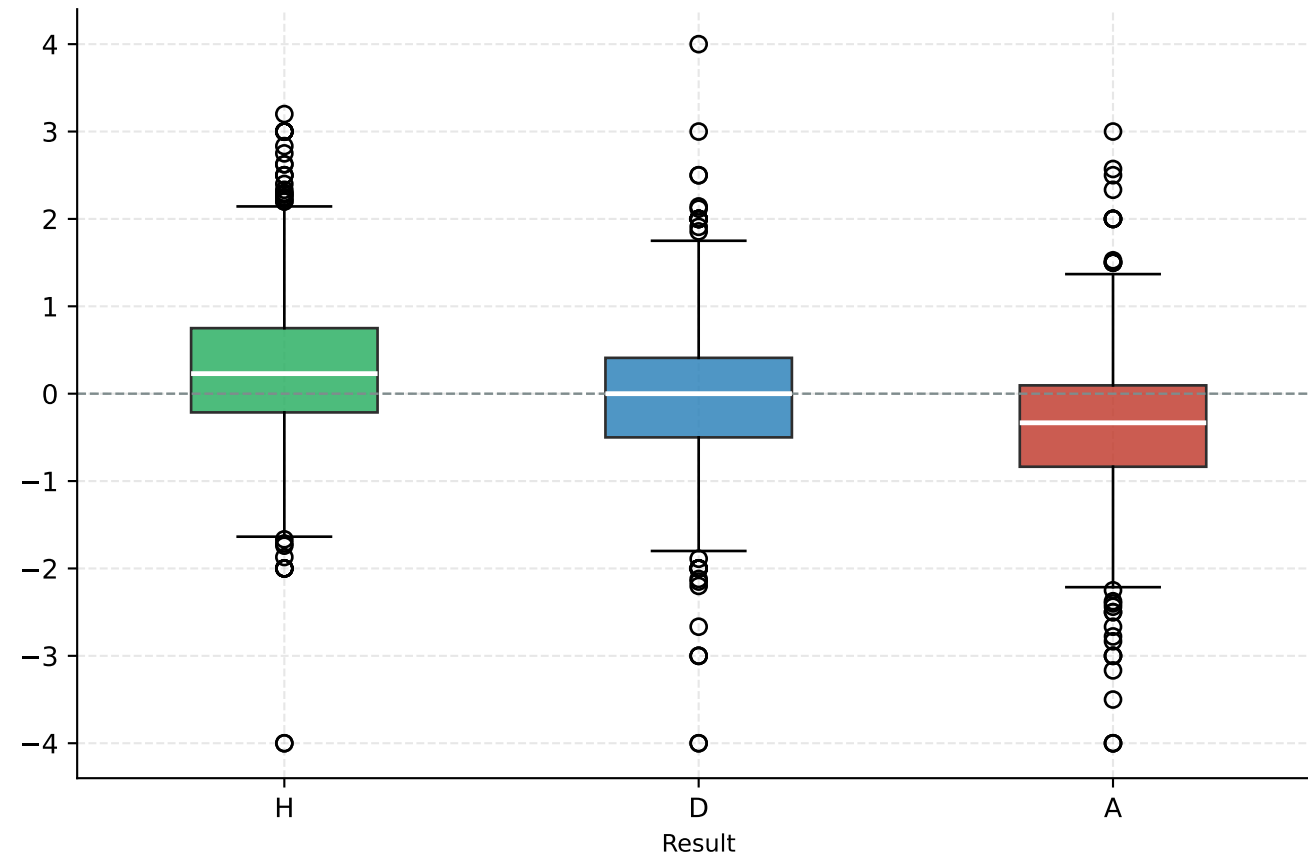


Away Team Position

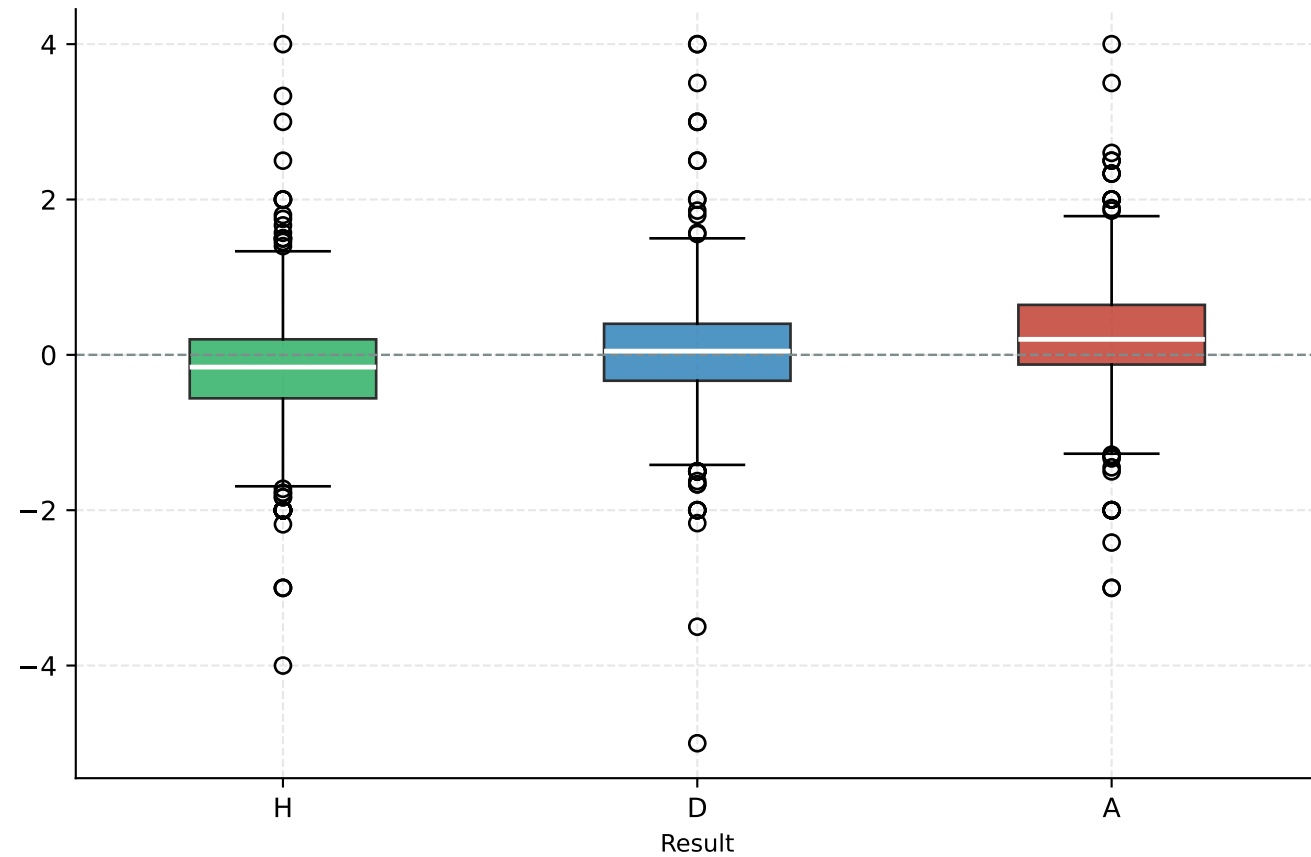


Derived Goal Differentials vs Match Result

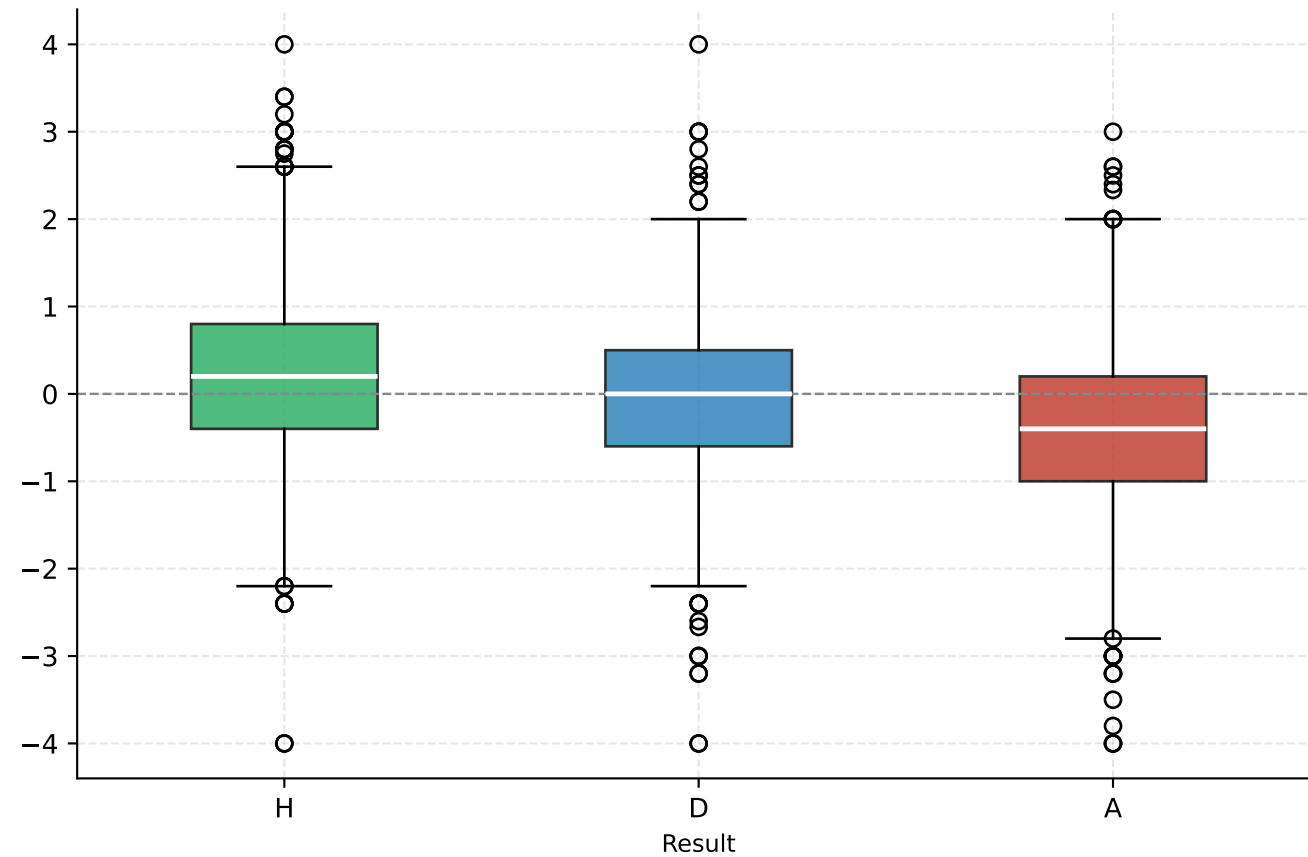
Avg Goals Scored Differential (Home – Away)



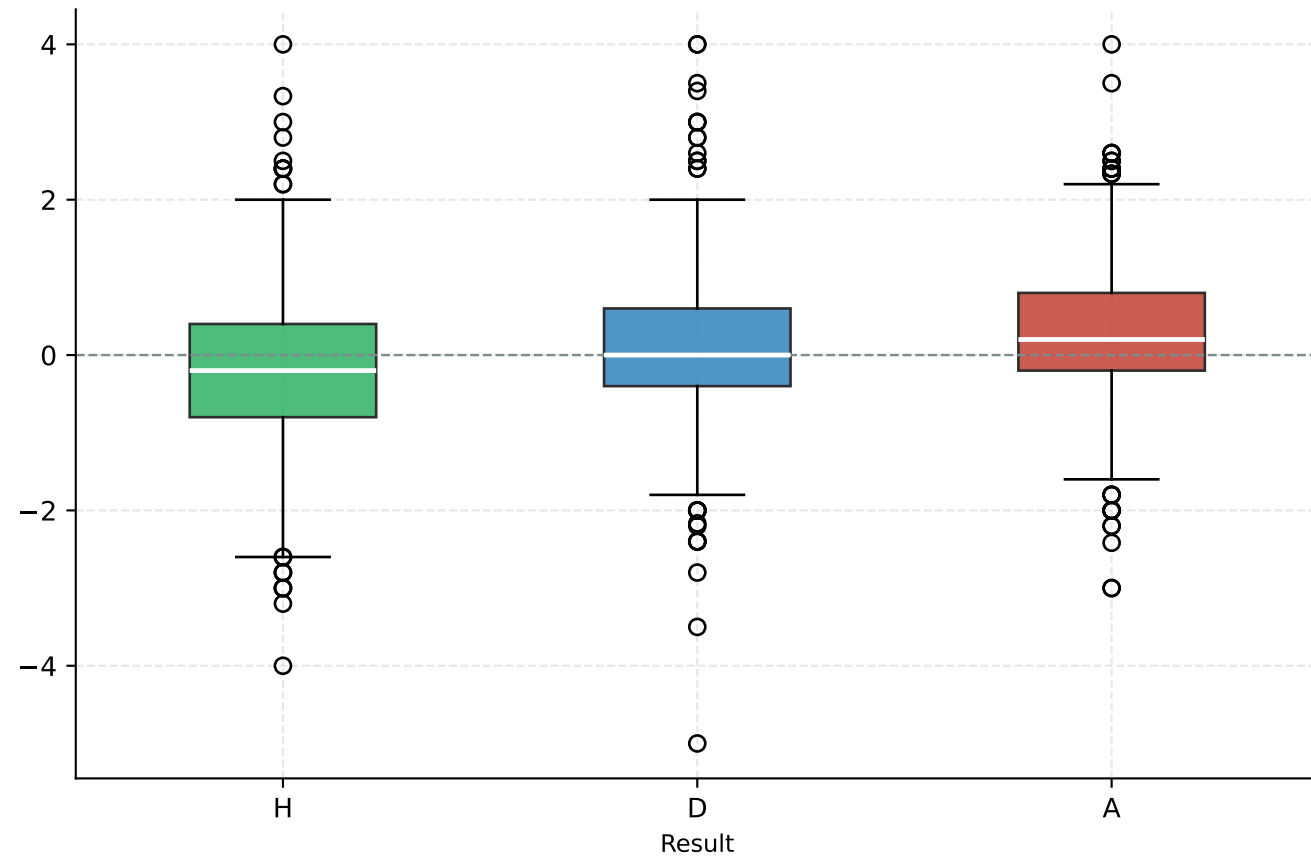
Avg Goals Conceded Differential (Home – Away)



Form Goals Scored Differential (Home – Away)

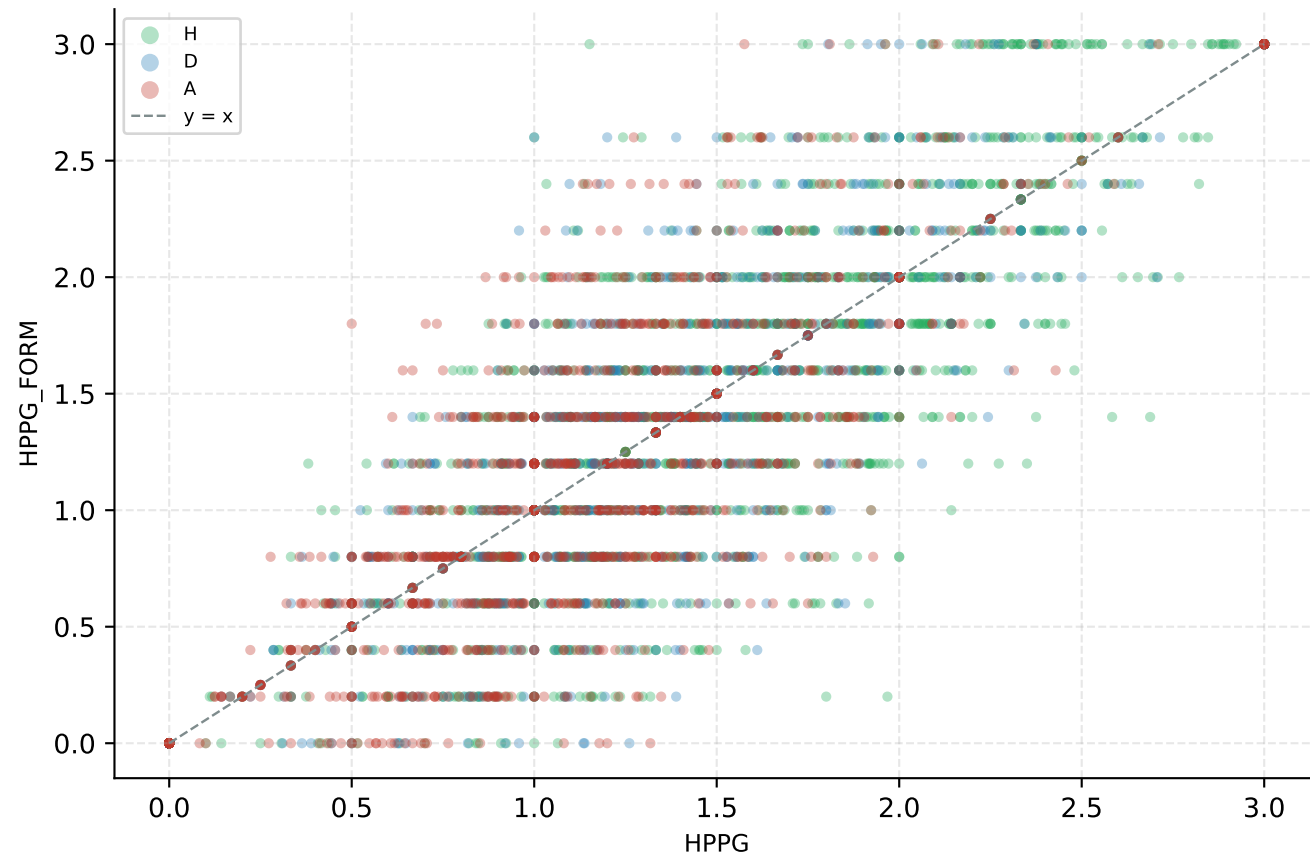


Form Goals Conceded Differential (Home – Away)

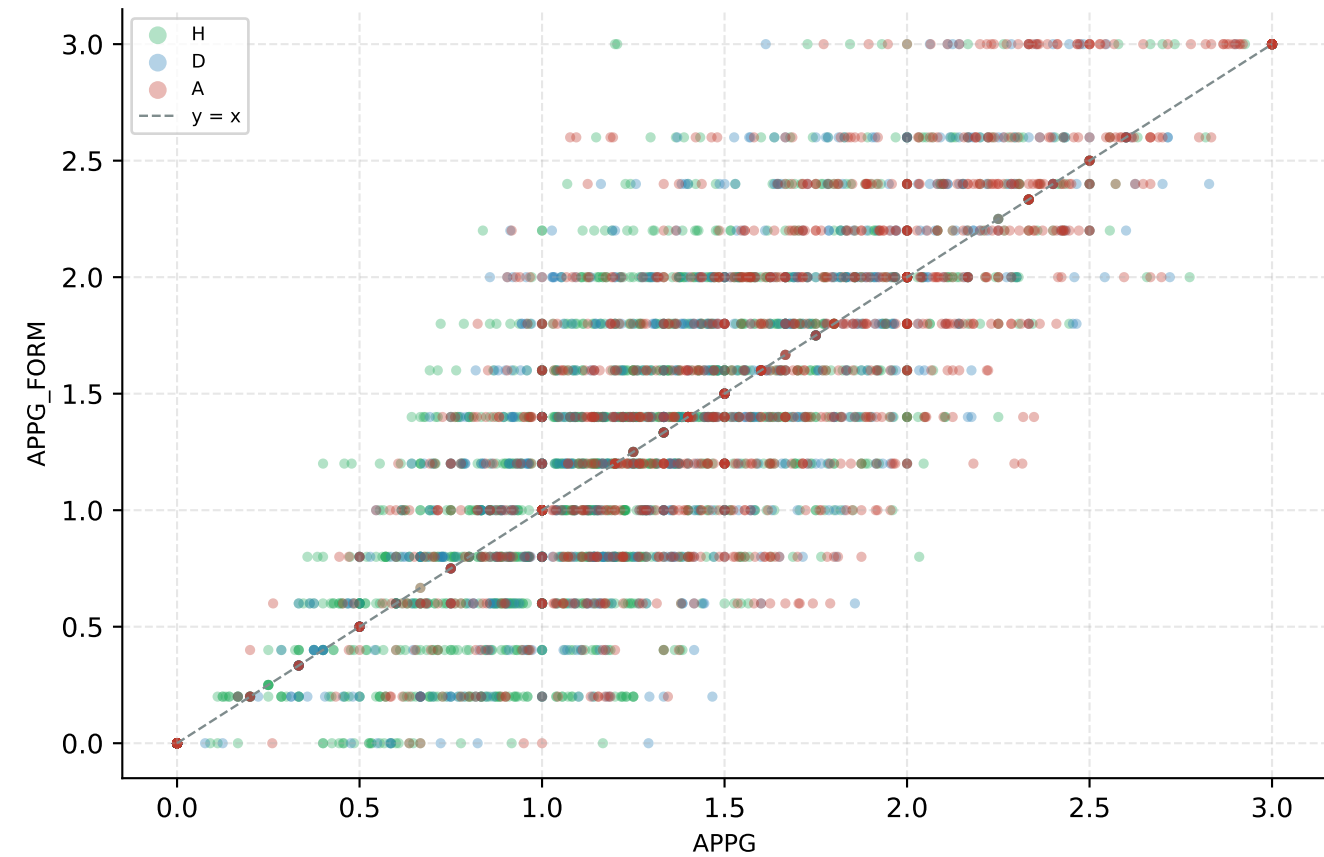


Season Average vs Recent Form (coloured by FTR)

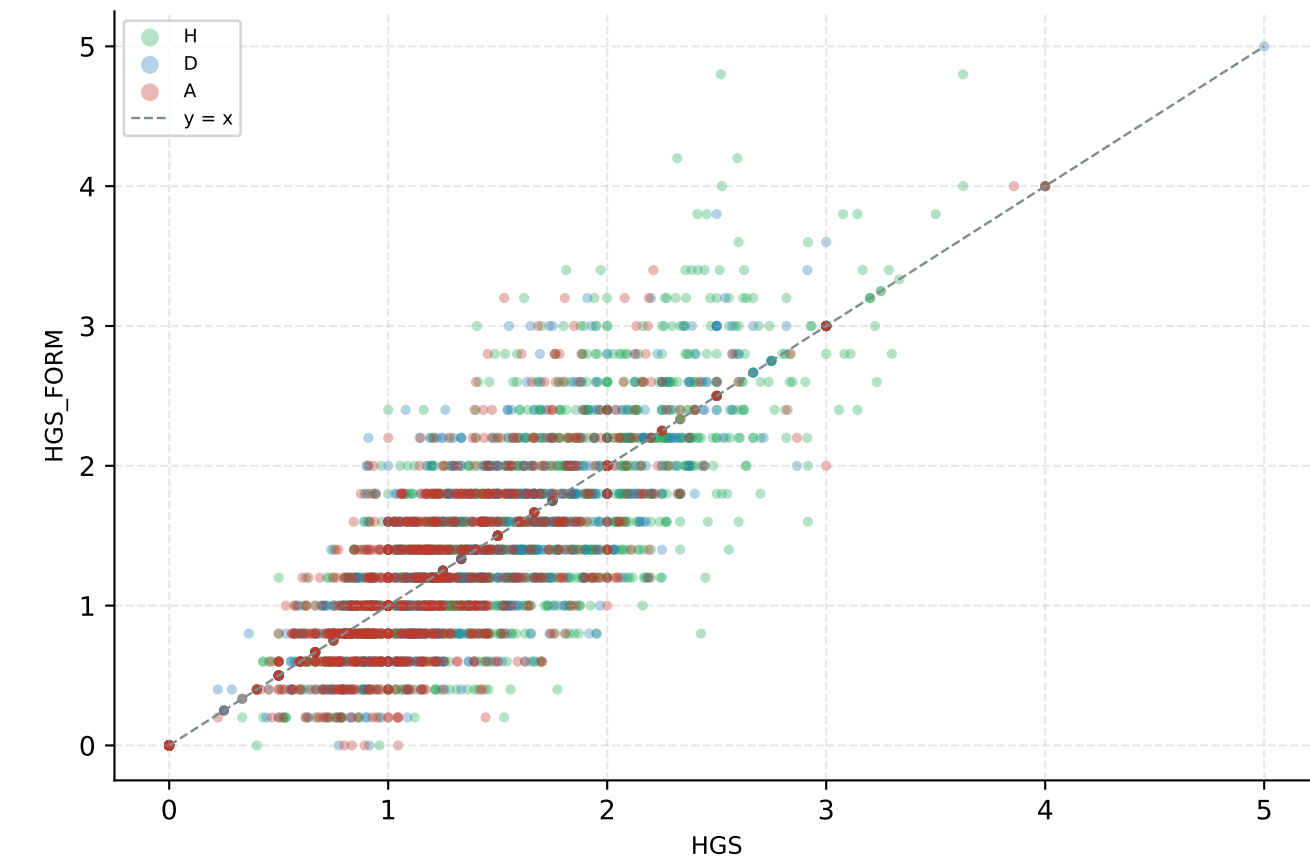
Home Points: Season vs Form



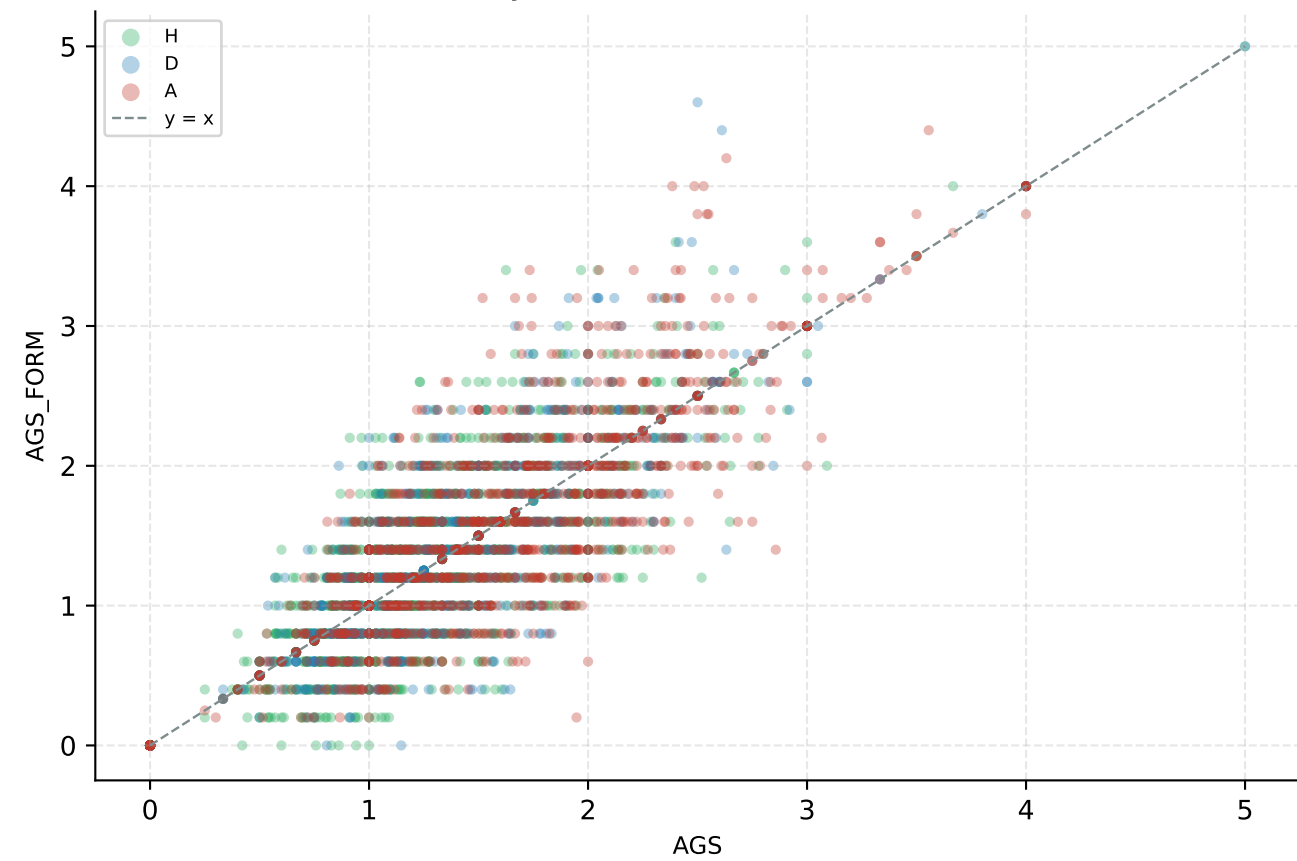
Away Points: Season vs Form



Home Goals Scored: Season vs Form



Away Goals Scored: Season vs Form



Shots & Discipline Form Profiles by FTR

